



A Real-Time Explainable Platform for Employee Attrition Prediction and Sentiment Analysis

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Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

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05/08/2026

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Abstract

Employee attrition is an ongoing and costly problem for modern organisations, because losing experienced staff drives up recruitment and training costs, disrupts operations and takes organisational knowledge with it. Previous studies have applied machine learning either to predict attrition or to analyse employee sentiment, but most of them offer little explanation of their predictions, handle class imbalance poorly, and do not bring structured employee records together with the text employees actually write. This project responds to those gaps with a real time explainable deep learning platform for employee attrition prediction and employee sentiment analysis.

The framework is built as two parallel pipelines. The attrition pipeline uses two modern tabular deep learning models, an FT-Transformer and a Tabular ResNet, implemented in PyTorch and trained on the IBM HR Analytics dataset of 1,470 records. The sentiment pipeline uses a BiLSTM and a CNN-BiGRU model with attention, implemented in TensorFlow and Keras and trained on Glassdoor employee reviews. Hypothesis testing with t tests and chi square tests identified 21 significant features from 30, and a configurable switch allowed every model to be trained twice, once on the complete feature set and once on the significant subset, so the effect of feature selection could be measured rather than assumed. SHAP, LIME, attention visualisation and permutation importance were used to explain both pipelines during analysis, while the deployed application explains each live prediction by masking, so that in every case a prediction can be traced back to the features or words behind it.

The FT-Transformer reached a test accuracy of 87.76% on attrition, and the CNN-BiGRU model with attention reached 91.79% with a ROC-AUC of 0.972 on sentiment. Feature selection made almost no difference to accuracy, with the FT-Transformer moving from 87.76% to 87.07% and the Tabular ResNet holding at 85.03%, although it lifted the ROC-AUC of the Tabular ResNet from 0.740 to 0.812 and raised lever precision from 0.69 to 0.75. The clearest limitation was class imbalance, with only around 16% of employees leaving, recall on the Leave class stayed at 0.39 for the FT-Transformer and 0.30 for the Tabular ResNet, so headline accuracy overstates what the models can do on the cases that matter most. Cross dataset validation on the Job Change and Employee Reviews data produced attrition accuracies near 80% and sentiment accuracies above 91%, which confirms the models learned transferable patterns rather than memorising one dataset. Across every explainability method, OverTime was the single most consistent driver of attrition, ranking highly under SHAP, under FT-

Transformer attention and under Tabular ResNet permutation importance, with Department, MonthlyIncome, EnvironmentSatisfaction and YearsSinceLastPromotion supporting it. On the text side, reviews mentioning stress, workload and weak support were the strongest negative signals, while stability and supportive management drove positive ones.

All four models were deployed as an interactive Gradio application on Hugging Face Spaces, with the user choosing the architecture on each tab. Because of the recall problem above, the attrition tab is served by the Job Change models rather than the IBM ones, so that live predictions are not skewed by the imbalance of the original dataset. Each prediction returns a plainly worded verdict, a probability, and a masking based explanation of the features or words that drove it. The result is a single deployable system that combines attrition prediction, sentiment analysis and explanation, which the reviewed literature does not do.

Key words: Employee Attrition Prediction, Employee Sentiment Analysis, Explainable AI, FT-Transformer, CNN-BiGRU, SHAP, LIME, Hugging Face Spaces

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Chapter 1 Introduction

Employee attrition refers to the process by which employees leave an organisation, whether voluntarily or involuntarily [1]. High levels of attrition present a significant challenge for modern organisations because replacing experienced staff increases recruitment and training costs, reduces productivity, disrupts everyday operations and results in the loss of valuable knowledge and expertise. The consequences extend beyond the organisation itself. Human resources departments and managers must invest considerable time and resources in recruiting, onboarding and training replacements, while the remaining staff often experience heavier workloads, reduced morale and lower overall efficiency.

Recent advances in artificial intelligence and deep learning have made it possible to apply predictive analytics to workforce management [3]. By analysing employee attributes such as job role, salary, years of experience, job satisfaction and work life balance, deep learning models can estimate the likelihood that an employee will leave or not. In parallel, employee reviews and free text feedback offer valuable insight into satisfaction, workplace culture and organisational problems [2]. Sentiment analysis can automatically classify such reviews as positive or negative, which helps organisations understand employee opinion at scale.

Although earlier studies have applied machine learning algorithms such as Random Forest, K-Nearest Neighbours and Naive Bayes to attrition prediction, many of these approaches suffer from class imbalance, limited optimisation, weak feature selection and a lack of interpretability. Most studies also address either attrition prediction or sentiment analysis in isolation, without integrating the two into a single decision support system. As a result, HR professionals often receive predictions without any explanation of the factors behind them, which makes it difficult to take appropriate retention actions.

To address these limitations, this project presents a real time explainable deep learning platform for employee attrition prediction and sentiment analysis. The platform combines two pipelines. Modern tabular deep learning models predict attrition from structured HR data, while recurrent and convolutional neural networks analyse the sentiment of employee review text. Explainable artificial intelligence techniques, including SHAP and LIME, are applied to highlight the features and words that most influence each prediction. The trained models are deployed as an interactive web application using Hugging Face Spaces, which enables decision makers to submit employee information or review text and receive instant predictions together with meaningful explanations.

1.1 Problem Description, Context and Motivation

Employee turnover is a major problem for organisations of all sizes and across all industries. When experienced employees leave, organisations spend more on hiring and training, and overall productivity declines. Many organisations also find it difficult to identify in advance the employees who are most likely to leave, and to understand the reasons behind their decisions. The problem is frequently identified only after employees have already departed, by which point retention interventions are no longer possible.

The problem directly affects organisations, HR departments, managers and employees. High turnover increases the workload of remaining staff, reduces team performance and damages the overall working environment. It is important to solve because early and explainable identification of employees at risk allows managers to intervene before a resignation occurs. Sentiment from employees' feedback also gives a complementary and people-centred perspective on workforce health, in addition to structured data.

For this project, the motivation is to go beyond the predictions that cannot be explained, and to go beyond systems that treat structured data and employee opinion separately. The project's goal is to deliver an evidence-based and transparent solution to HR teams that will help them make proactive retention decisions by predicting attrition, analysing sentiment and explaining both in a single deployable platform.

1.2 Objectives

The objectives of the project are the tangible steps taken to achieve the aim described below.

1. To collect and prepare suitable employee attrition and employee review datasets for analysis.
2. To preprocess the data by handling missing values, encoding categorical features, scaling numerical features and preparing text data for sentiment analysis.
3. To perform feature engineering and statistical feature selection in order to identify the factors most strongly associated with attrition.
4. To develop deep learning models, specifically an FT-Transformer and a Tabular ResNet, for predicting attrition from structured data.
5. To develop deep learning models, specifically a BiLSTM and a CNN-BiGRU model with attention, for classifying the sentiment of employee reviews.

6. To compare model performance using the complete feature set versus a statistically selected significant feature set, in order to assess how feature selection affects prediction accuracy and efficiency.
7. To evaluate and compare the models using Accuracy, Precision, Recall, F1 score and ROC-AUC.
8. To apply explainable artificial intelligence techniques, including SHAP and LIME, to explain the predictions of the models.
9. To deploy the selected models as an interactive web application using Hugging Face Spaces.

1.3 Methodology

The project follows a quantitative and experimental research methodology in order to develop an explainable deep learning system for attrition prediction and sentiment analysis. The methodology consists of two parallel pipelines. The first predicts attrition from structured HR data and the second analyses sentiment from employee review text. The models are evaluated, explained using explainable artificial intelligence and finally deployed as an interactive application.

1.3.1 Design

The system is composed of two parts deep learning system. The first section deals with prediction of attrition based on structured data. It starts with data collection, followed by data understanding and data preprocessing, where missing values are filled in, categorical variables are converted to numerical, numerical variables are scaled, and class imbalance is examined. Features are then engineered and statistically tested for their ability to correlate to the attrition set, the FT-Transformer and Tabular ResNet models are then trained and compared. The second part is about the sentiment analysis based on the review text. The reviews are preprocessed to be cleaned, tokenized and transformed into numerical sequences before being used to train the BiLSTM and CNN-BiGRU models with attention for sentiment classification into positive and negative. After training all the models, they are tested, then explained in terms of SHAP and LIME, and finally selected models are deployed.

The design was selected since previous studies were based on traditional machine learning algorithms that tend to perform worse on complex and imbalanced HR data. Deep learning models were chosen because they are capable of learning complex relationships in structured

and textual data, SHAP and LIME were chosen to provide better explainability in the sense of providing explanations after training.

1.3.2 Testing and Evaluation

The datasets are split into training, validation and testing sets. The training subset is used to fit the model parameters, the validation subset is used to decide when to stop training, and the held out test subset is used to test final performances. All models are evaluated based on Accuracy, Precision, Recall, F1 score and ROC-AUC with the aid of confusion matrices and complete classification reports. Since the attrition datasets are usually imbalanced, the other metrics are used to evaluate the performance on the minority class, and are better than Accuracy. Both the complete feature set and the statistically significant feature subset are used to test models, to determine if feature selection enhances prediction.

1.3.3 Project Management

The project is developed in an iterative manner with each phase completed, tested and improved before the next phase is started. The process involves gathering data, preprocessing it, engineering features, building models, assessing their performance, enhancing their explainability and subsequently deploying them. Once each model is constructed, its performance is examined and changes to the configuration are made to the model, and the performance of the models are compared. This cycle of iteration will help ensure that the solution will be continuously improved and that it would be interpretable while maintaining predictive accuracy.

1.3.4 Technologies and Processes

Python is used as the main programming language for preprocessing, model development and evaluation. PyTorch is used to implement the FT-Transformer and Tabular ResNet models because it offers flexible and efficient support for deep learning on structured data. TensorFlow and Keras are used for the BiLSTM and CNN-BiGRU models because of their strong support for natural language processing. Scikit-learn is used for preprocessing, feature selection and evaluation metrics, while SHAP and LIME are used to explain predictions. The datasets are obtained from Kaggle, model development is carried out in Jupyter and Kaggle notebooks, and the final models are deployed using Hugging Face Spaces. These technologies were selected because together they provide reliable support for deep learning, natural language processing, explainable artificial intelligence and web based deployment.

1.4 Legal, Social, Ethical and Professional Considerations

Legal considerations: The datasets used in this project were obtained from Kaggle under their published open licences for academic research, and they have also been used in several peer reviewed studies. The datasets are publicly available and contain synthetic or anonymised records with no personally identifiable information, so no personal data was collected or processed by the author.

Social considerations: The platform is intended to help organisations understand and reduce attrition and to improve workplace conditions. It is designed to support HR decision making and should not be used to monitor individual employees unfairly or to make automated employment decisions without human oversight.

Ethical considerations: Because the project uses only publicly available secondary datasets and does not involve human participants, surveys or interviews, formal ethical clearance was not required. Explainability methods were applied to make the model's predictions more transparent and to reduce the risk of unclear or unaccountable decisions.

Professional considerations: The project followed good academic and professional practice by properly acknowledging the datasets, software tools and prior research used. The models were developed, tested and evaluated in a standard way with no dissembling of the results or limitations, where minority class performance did not improve the model was not discarded, but rather the results and limitations were reported.

1.5 Background

One of the key issues in human resource management is employee turnover, as it costs more to hire and train new employees and leads to lost productivity if experienced employees leave. Employees are one of the most valuable assets of an organisation, and it has become an important business objective to attract and retain employees, and organisations increasingly rely on data driven techniques to know and predict when employees are likely to leave [1].

Historical human resource data has been extensively used for attrition prediction with the help of machine learning. Typical algorithms for identifying likely leavers include Random Forest, K-Nearest Neighbours and Naive Bayes, but these algorithms are not well suited to imbalanced data sets and are not able to detect complex relationships between employee characteristics. These limitations were pointed out by Krishna, Dwivedi and Murti and it was recommended using deep learning and sentiment analysis to enhance the prediction [1].

There are also some common limitations in the existing systems of attrition. Even though many fail to deal with class imbalance, do not conduct feature selection, or explain their predictions, or compare a number of models, it will be hard for HR personnel to rely on the results or apply them. There is recent research that argues that explainable AI techniques, like SHAP and LIME, can enhance transparency and aid in better decision-making [3] [4]. In the wake of these observations, the present project proposes an explainable deep learning framework that integrates structured attrition prediction with sentiment analysis of employee reviews for more accurate, transparent and practical decision support.

1.6 Structure of Report

This report continues in the following organisation. The results are summarized in Chapter 2, where the literature review is presented and the strengths and weaknesses of the candidate technologies are explained, as well as the research gaps that justify this project. Chapter 3 presents the system design, data sources, exploratory analysis, preprocessing, architectures of the four deep learning models, explainability methods and deployment artefacts. The evaluation results for both pipelines are presented and discussed in Chapter 4, with the full feature set and a second HR dataset for cross dataset evaluation and compared to related work. The report closes with a summary of how well the goals were achieved and what work remains to be done, followed by a critical examination of the project undertaken in Chapter 5.

Chapter 2 Literature – Technology Review

This chapter summarizes the literature on employee attrition prediction and sentiment analysis, discusses the technologies that may be used to solve the problem and highlights the research gaps that the project aims to fill. The literature review looks at the current approaches and the constraint of those approaches, the technology review justifies the technology that has been looked at in this project and the final comments explain how the findings lead to the methodology.

2.1 Literature Review

Employee attrition is the term used when employees leave an organisation either by choice or by force. Human resource management is a significant problem because the replacement of experienced employees requires a higher recruitment and training cost, it causes disruption to the production process and the loss of knowledge and skills. In response, many of the researchers have devised machine learning models to predict the attrition and in recent years some have considered the sentiment of the employee reviews to understand the satisfaction. This review presentation will discuss the key strategies, their strengths and limitations, and areas that this project will address.

Most of the earlier studies rely on the IBM HR Analytics Employee Attrition dataset, which consists of 1470 rows and 35 attributes. The authors of this project, Krishna, Dwivedi and Murti [1] compared the performance of the KNN, Naive Bayes and weighted ensemble models, with the Random Forest model which gave an accuracy range of 75% to 84%. Both Rohit et al. [2] and Inggianti and Mulyana [3] achieved the best results with Random Forest with accuracy of 87.46% and 87.41% respectively, and Pathak et al. [5] reported an accuracy of 86.2% with XGBoost. The results indicate that traditional models give reasonable performance, but that different algorithms can achieve comparable performance on the same data, meaning that there is not a significant benefit to be gained by switching algorithms.

A recurring problem in this body of work is class imbalance. In the IBM dataset only around 16 per cent of employees actually leave, so a model can achieve high overall accuracy simply by predicting that most employees stay, while failing to identify the minority who leave. Krishna et al. [1] reported 84% accuracy yet correctly identified only a small number of leavers, and Inggianti and Mulyana [3] reported many false negatives. These findings show that

Accuracy alone is inadequate for this task, and that Precision, Recall, F1 score and ROC-AUC give a more reliable picture of performance on the minority class.

Deep learning has been applied less often to attrition. Ramya and Sanjay [4] built a hybrid CNN-LSTM model by reshaping structured data into a sequential form. Although it reached 83 per cent accuracy, its ROC-AUC of only 0.54 was barely better than random and lower than the 0.70 achieved by their simpler voting classifier, which suggests that CNN-LSTM architectures are not well suited to structured tabular data. None of the reviewed studies used modern tabular deep learning models such as the FT-Transformer or the Tabular ResNet [8], which use attention and residual connections to model feature interactions and have shown strong results on tabular benchmarks. This represents a clear research gap.

A further limitation is the lack of explainability. Most studies explain their models using only Random Forest feature importance [2], [3], which gives a global overview but cannot explain an individual prediction. Pathak et al. [5] proposed a Logic Tensor Network that combines learning with logical rules in order to improve interpretability, but it was computationally expensive, required careful tuning and depended on manually defined rules, which makes it difficult to apply in practice. This creates an opportunity for model agnostic methods such as SHAP [9] and LIME [10], which explain individual predictions without altering the model.

Sentiment analysis of employee opinion has also been studied, since reviews capture information that structured data cannot. Rahman et al. [7] showed that combining transformer models with recurrent networks improves classification, with their RoBERTa-BiLSTM model reaching 92.36% accuracy on IMDb and 80.74% on the Twitter US Airline dataset, and they found that augmentation helps when data are imbalanced. Reddy et al. [6] used BiLSTM and RoBERTa-BiLSTM and DistilBERT-GRU with up to 90% accuracy and explained with LIME. The studies, however, were conducted on movie reviews and social media feedback as opposed to workplace feedback, and it was uncertain whether their models would work on real feedback.

Two research gaps emerge from this review. First, most attrition studies rely on traditional machine learning, do not effectively address class imbalance, focus mainly on Accuracy and provide only limited explainability, which points towards modern tabular deep learning, better evaluation and explainable artificial intelligence. Second, none of the reviewed studies combine attrition prediction, sentiment analysis of employee reviews, explainability and deployment into a single decision support system. Krishna et al. [1] explicitly recommended future work along these lines, and the present project responds directly by combining the FT-Transformer and

Tabular ResNet for attrition, the BiLSTM and CNN-BiGRU with attention for sentiment, SHAP and LIME for explanation, and deployment as an interactive application.

Limitations of the existing pipeline: A structured comparison of the pipelines reported in prior work reveals a consistent set of missing stages. These include the absence of thorough data understanding, data quality checks, class imbalance analysis, data augmentation, feature engineering and selection, explainability, cross validation, error analysis and model comparison. Each omission is small in itself but, collectively, they can result in lower accuracy, predictions of the majority class, poorer generalisation and limited interpretability. Each of these steps is specifically included in the methodology outlined in Chapter 3.

2.2 Technology Review

This section reviews the technologies considered for the project and explains the rationale for each choice, covering the models for structured data, the models for review text, the explainability techniques and the deployment platform.

2.2.1 Modelling Structured Employee Data

Several technologies can predict attrition from structured HR data. They include traditional models such as Random Forest and XGBoost, sequential deep learning models such as CNN-LSTM, and modern tabular deep learning models such as the FT-Transformer, TabNet and the Tabular ResNet [8]. Traditional models perform well on small tabular datasets and dominate the existing literature, but as the literature review showed they struggle to identify likely leavers and yield only marginal improvements. CNN-LSTM models were also considered but are designed primarily for sequential data and are not well suited to structured records. For these reasons the FT-Transformer and the Tabular ResNet were selected. The former uses self attention to model relationships between features, while the latter uses residual connections to train deeper networks effectively. Both are purpose built for tabular data and were implemented in PyTorch for its flexibility and efficient training.

2.2.2 Modelling Employee Review Text

Sentiment analysis can be approached with TF-IDF combined with classifiers such as Support Vector Machines, with sequence models such as LSTM, BiLSTM and GRU, or with transformer models such as BERT and RoBERTa [6], [7]. Transformer models typically achieve very high accuracy but demand more computation and longer training, which is a drawback for a project that also aims to deliver a responsive web application. Models offering

a better balance of accuracy and efficiency were therefore preferred, and the BiLSTM and the CNN-BiGRU with attention were selected. The BiLSTM reads each review in both directions in order to capture context, while the CNN-BiGRU model first extracts local phrase patterns with convolutional layers, then models sequence context with bidirectional GRUs, and finally uses attention to focus on the words most relevant to sentiment. Both were implemented in TensorFlow and Keras.

2.2.3 Explainability

Various explainability approaches were discussed. Some methods, like Logic Tensor Networks, embed interpretability into the model, and some, like SHAP and LIME, provide explanations of predictions after training [5], [9], [10]. This project chose these two models, SHAP and LIME, because they are model agnostic, and explain the predictions of different architectures without changes. SHAP is used to find the most important employee attributes for an attrition prediction, whereas LIME is used to find the most important words for a sentiment classification. They work together to enhance transparency and enable HR professionals to grasp the rationale behind each result.

2.2.4 Deployment and Supporting Tools

Various deployment methods like Streamlit and cloud platforms were evaluated. Hugging Face Spaces was chosen as it offers a free cloud hosting solution and allows users to run the app on a web browser without installing anything. Python is the primary language for the wider tool chain, with scikit-learn for metrics and preprocessing, PyTorch for attrition models and TensorFlow/ Keras for sentiment models. All the development was done in Jupyter and Kaggle notebooks and all datasets were retrieved from Kaggle.

2.3 Summary

The literature review revealed that the majority of these previous attrition studies relied on the classic machine learning models on the IBM HR Analytics dataset with good overall accuracy but failed to correctly predict the attrition, due to the class imbalance. Most focus on accuracy, apply limited explainability and do not use modern tabular deep learning models. Only movie reviews or social media data was used to analyze sentiment in the reviewed studies and no studies used both attributes to predict attrition in one explainable system.

The technology review addressed the selection of the FT-Transformer and Tabular ResNet for the purpose of predicting attrition, of the BiLSTM and CNN-BiGRU with attention for

sentiment analysis, of the SHAP and LIME methods for explaining the models, and of Hugging Face Spaces for deployment. The overall findings form a methodology, which addresses class imbalance, compares various deep learning models, assesses model performance using a complete set of metrics, provides a transparent explanation of predictions, and integrates both pipelines into an integrated deployable decision support tool for HR.

Chapter 3 Implementation

This chapter describes the practical implementation of the real time explainable platform, covering the two pipelines introduced in Chapter 1. The first is an attrition prediction pipeline built on the IBM HR Analytics dataset and the second is a sentiment analysis pipeline built on Glassdoor employee reviews. The chapter presents the system design, the data sources, the exploratory data analysis, the preprocessing and feature engineering, the architectures and training of the four deep learning models, the explainability methods and the deployment artefacts, before summarising the work that leads into the evaluation chapter.

3.1 System Design

The system is organised as two parallel pipelines that share a common development workflow but use different frameworks and operate on different data modalities. The first pipeline transforms structured HR records through data understanding, preprocessing, statistical feature selection and model training with the FT-Transformer and the Tabular ResNet. The second pipeline transforms free text reviews through cleaning, tokenisation and model training with the BiLSTM and the CNN-BiGRU model with attention. Both pipelines converge on a shared evaluation, explainability and deployment stage, in which the selected models are exported together with their preprocessing objects and served through a Hugging Face Spaces web application. This design ensures that the two pipelines can be developed and evaluated independently while presenting a single and coherent interface to the end user as Figure 1 shows below.

My proposed Pipeline:

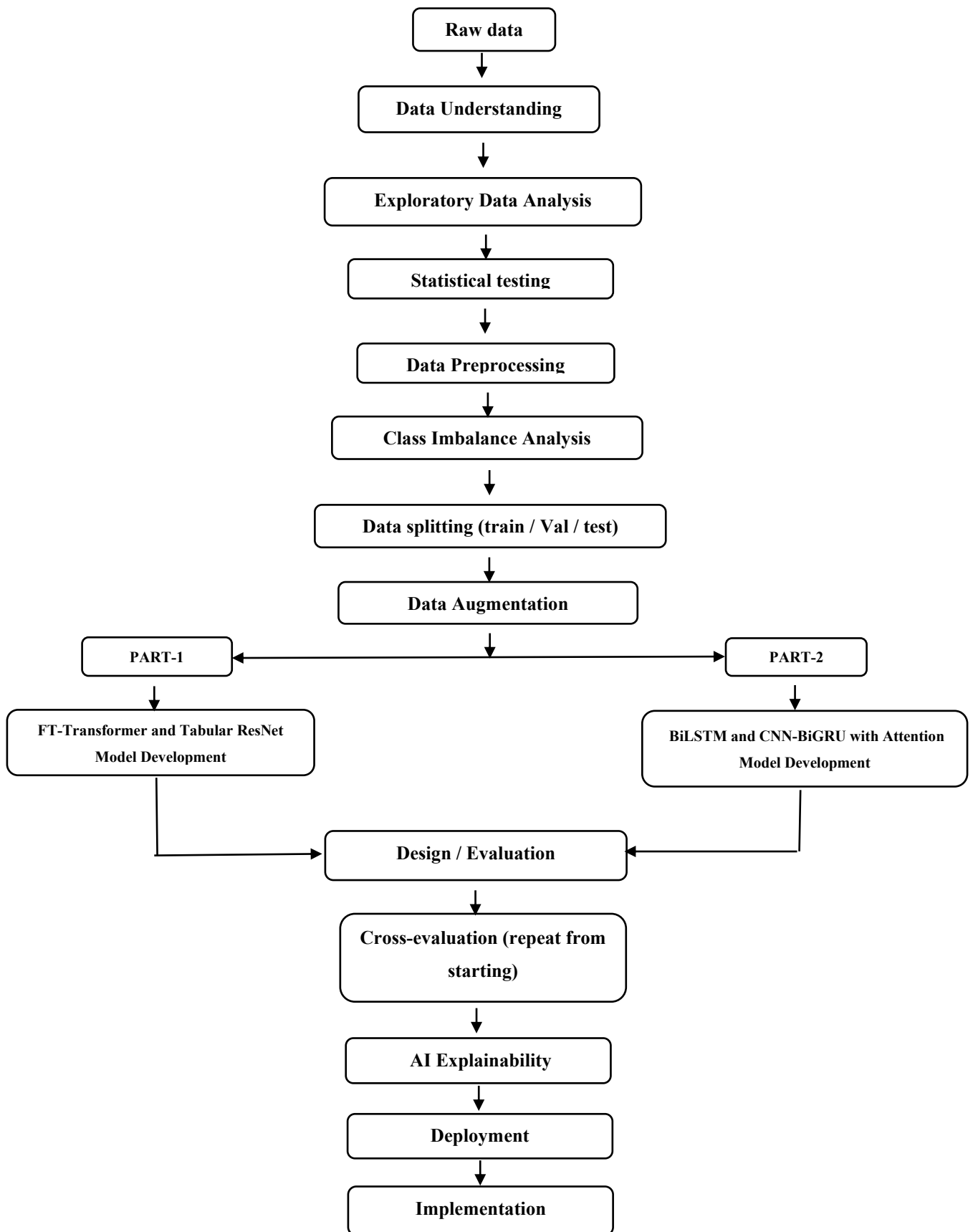


Figure 1 Proposed pipeline for the expandable employee attrition prediction and sentiment analysis platform

3.2 Data Sources

The project uses four publicly available secondary datasets obtained from Kaggle, each supporting one pipeline. Both were selected deliberately rather than arbitrarily, for the reasons set out below.

3.2.1 IBM HR Analytics Employee Attrition Dataset

There are 1,470 employee records in the IBM HR Analytics dataset, and the data is represented by 35 attributes. These include satisfaction measures like job satisfaction, environment satisfaction, work life balance, and job related information including job role, department, job level and monthly income, and demographic information like gender and marital status, age, etc. The target variable Attrition is binary and represents employee attrition or not. Three reasons were given for choosing this data set. It is the most consistently used dataset in previous published studies, making it easy to compare with previous benchmarks. While it is a small number of records, it has a lot of features making it suitable for understanding whether more value can be extracted from the same features using modern tabular deep learning models. Finally, it is a synthetic dataset released for research, so it contains no personally identifiable information.

3.2.2 Glassdoor Job Reviews Dataset

The Glassdoor Job Reviews dataset is considerably larger, containing 838,566 reviews described by 20 columns. These include free text fields for employees' likes and dislikes of their employer as well as a star rating from 1-5 stars. This dataset was selected because the literature on sentiment and sentiment analysis is dominated by movie reviews and social media posts, the vocabulary, tone and topics of which are not representative of real workplace feedback. The Glassdoor data directly fills this gap and the size of the data set is large enough to avoid overfitting when training deep learning text models with a small vocabulary.

3.2.3 Job Change of Data Scientists Dataset

The dataset Job Change of Data Scientists consists of 19,158 observations and 14 attributes. These include demographic and background information like gender, level of education, main discipline and relevant experience, and company related information like company size, company type. The target variable will determine if a candidate is seeking a job change. For the task of attrition, this data set was used as a cross-validation set. It enables the models to be evaluated in a fresh domain and feature structure other than the IBM data, allowing the models

to be evaluated for their generalisation performance over data not used in their training. It is publicly available without any personally identifiable information and was obtained from similar sources as the primary dataset.

3.2.4 Employee Reviews Dataset

The Employee Reviews dataset consists of 67,529 reviews with 19 columns. These include free text review fields and a user rating, like the Glassdoor data, but from a different source. This data set served as a cross validation set for sentiment task. It examines if the sentiment models generalize to workplace feedback in different settings beyond the ones they have been trained on. It is large enough to be a meaningful independent test of model accuracy.

3.3 Data Understanding and Exploratory Data Analysis

Both datasets were first analyzed thoroughly to gain insight into their structure, quality and patterns, as building models on poorly understood data can introduce hidden quality issues and unreliable predictions. The IBM data set was examined to verify the number of rows and columns and the type of data in each field. This showed a combination of numeric fields like Age, MonthlyIncome, and TotalWorkingYears, and categorical fields like JobRole, Department, and OverTime, with the target field showing as the text values Yes and No. The same inspection was used for the Glassdoor dataset, with the exception that the free text review fields were also included as well as the star rating column which was subsequently transformed to a sentiment label. Here, detailed exploratory analysis is provided only for the IBM dataset and the Glassdoor dataset, which are the main ones used for model development. The same analysis of structure, quality and patterns was applied to the cross validation datasets, but without reporting all the results here to avoid repetition, the datasets simply provide independent checks of generalisation.

3.3.1 Descriptive Statistics

Table 1 summarises the descriptive statistics of the numeric features in the IBM dataset. MonthlyIncome, YearsAtCompany and YearsSinceLastPromotion are some of the features characterized by a strong right skew, meaning that a few employees have high values while the majority have lower values. These observations are confirmed by the skewness and kurtosis statistics in Table 2, which show that the features such as Age and DailyRate have near symmetric distributions.

Table 1 Descriptive statistics of the numerical features in the IBM HR Analytics dataset

Feature	Count	Mean	Std	Min	25%	Median	75%	Max
Age	1470	36.92	9.14	18	30	36	43	60
DailyRate	1470	802.49	403.51	102	465	802	1157	1499
DistanceFromHome	1470	9.19	8.11	1	2	7	14	29
Education	1470	2.91	1.02	1	2	3	4	5
EnvironmentSatisfaction	1470	2.72	1.09	1	2	3	4	4
HourlyRate	1470	65.89	20.33	30	48	66	83.75	100
JobInvolvement	1470	2.73	0.71	1	2	3	3	4
JobLevel	1470	2.06	1.11	1	1	2	3	5
JobSatisfaction	1470	2.73	1.10	1	2	3	4	4
MonthlyIncome	1470	6502.93	4707.96	1009	2911	4919	8379	19999
MonthlyRate	1470	14313.10	7117.79	2094	8047	14235	20461	26999
NumCompaniesWorked	1470	2.69	2.50	0	1	2	4	9
PercentSalaryHike	1470	15.21	3.66	11	12	14	18	25
PerformanceRating	1470	3.15	0.36	3	3	3	3	4
RelationshipSatisfaction	1470	2.71	1.08	1	2	3	4	4
StockOptionLevel	1470	0.79	0.85	0	0	1	1	3
TotalWorkingYears	1470	11.28	7.78	0	6	10	15	40
TrainingTimesLastYear	1470	2.80	1.29	0	2	3	3	6
WorkLifeBalance	1470	2.76	0.71	1	2	3	3	4
YearsAtCompany	1470	7.01	6.13	0	3	5	9	40
YearsInCurrentRole	1470	4.23	3.62	0	2	3	7	18
YearsSinceLastPromotion	1470	2.19	3.22	0	0	1	3	15
YearsWithCurrManager	1470	4.12	3.57	0	2	3	7	17

Table 1 shows that the numeric features are distributed unevenly. The age range is from 18 to 60 with a mean age of approximately 37, meaning that a majority of workers are in the early to middle part of their careers. MonthlyIncome doesn't follow that pattern, it ranges from just over 1,000 to nearly 20,000, with the median coming in at 4,919, and the mean at 6,503, indicating that there are fewer high earners that are raising the average. The same is true for the tenure

fields. Around half of the employees have been with the company for about 5 years, with the max being 40, and YearsSinceLastPromotion is zero for at least a quarter of employees, and up to 15 for others. The ordinal survey items, such as JobSatisfaction, EnvironmentSatisfaction and WorkLifeBalance, stay within their fixed range and cluster around a value of three, which suggests that most employees give middling to slightly positive ratings. Compared to the other two columns such as Pay and Tenure, PerformanceRating has virtually nonexistent width variation, ranging from 3 to 4.

Table 2 Distribution characteristics of the numerical features

Feature	Skewness	Kurtosis	Interpretation
Age	0.413	-0.404	Slightly right skewed and approximately symmetric
DistanceFromHome	0.958	-0.225	Moderately right skewed as most employees live closer to work
Education	-0.290	-0.559	Slightly left skewed as higher education levels dominate
EnvironmentSatisfaction	-0.322	-1.203	Slightly left skewed as higher satisfaction is reported
JobInvolvement	-0.498	0.271	Slightly left skewed as higher involvement is common
JobLevel	1.025	0.399	Right skewed as more employees sit at lower job levels
JobSatisfaction	-0.330	-1.222	Slightly left skewed as satisfaction is generally higher
MonthlyIncome	1.370	1.005	Strong right skew driven by a few very high incomes
NumCompaniesWorked	1.026	0.010	Right skewed as most worked for fewer companies
PercentSalaryHike	0.821	-0.301	Moderately right skewed as most hikes are lower
PerformanceRating	1.922	1.696	Highly right skewed as few employees score highly
RelationshipSatisfaction	-0.303	-1.185	Slightly left skewed as higher responses dominate
StockOptionLevel	0.969	0.365	Right skewed as most hold lower stock option levels
TotalWorkingYears	1.117	0.918	Right skewed as fewer employees have long experience
TrainingTimesLastYear	0.553	0.495	Moderately right skewed as training sessions are few
WorkLifeBalance	-0.552	0.419	Slightly left skewed as better balance is reported
YearsAtCompany	1.765	3.936	Highly right skewed with long serving outliers
YearsInCurrentRole	0.917	0.477	Right skewed as shorter role durations are common
YearsSinceLastPromotion	1.984	3.613	Highly right skewed as many are not recently promoted
YearsWithCurrManager	0.833	0.365	Moderately right skewed as manager tenure is shorter

The values in Table 2 put numbers on what the ranges already hinted at. Both measures are approximately symmetric and well behaved, as Age and DailyRate are near zero. Pay and Tenure are inverse. MonthlyIncome, YearsAtCompany and YearsSinceLastPromotion are all skewed to the right with a positive skew greater than 1, and YearsAtCompany also has a high kurtosis of 3.94, indicating a few long-serving employees form a heavy right tail instead of a smooth curve. The satisfaction and involvement items have a slight negative skew as employees generally select the higher end of the scale. PerformanceRating is the most uneven of all, as almost everyone has the same rating and only a few people score higher. These patterns are the reason a standard scaler was later applied to the numeric features, as leaving the skewed pay and tenure columns on their original scales would allow them to dominate the models during training.

3.3.2 Univariate Analysis

Univariate analysis examined each variable in isolation. Although all variables were assessed through univariate, bivariate and multivariate analysis, only the most relevant and informative visualisations are presented here for brevity. Figure 2 shows the distribution of the target variable and confirms the substantial class imbalance in the IBM dataset, with far more employees staying than leaving. This imbalance is the single most important property of the dataset for later modelling, because it means a model can appear accurate while learning almost nothing about the employees who leave.

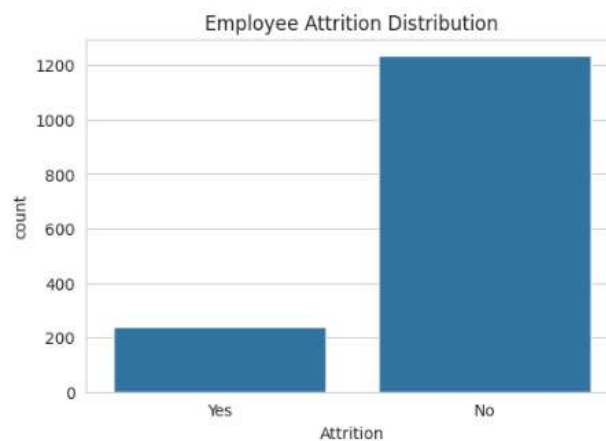


Figure 2 Target variable distribution

Figure 3 shows the distribution of the sentiment labels and overall star ratings derived from the Glassdoor reviews. Both distributions lean towards positive feedback, with four star and five star ratings dominating the sample before any balancing is applied. This mirrors the imbalance

seen in the attrition data and explains why the review data was later resampled into an equal number of positive and negative examples.

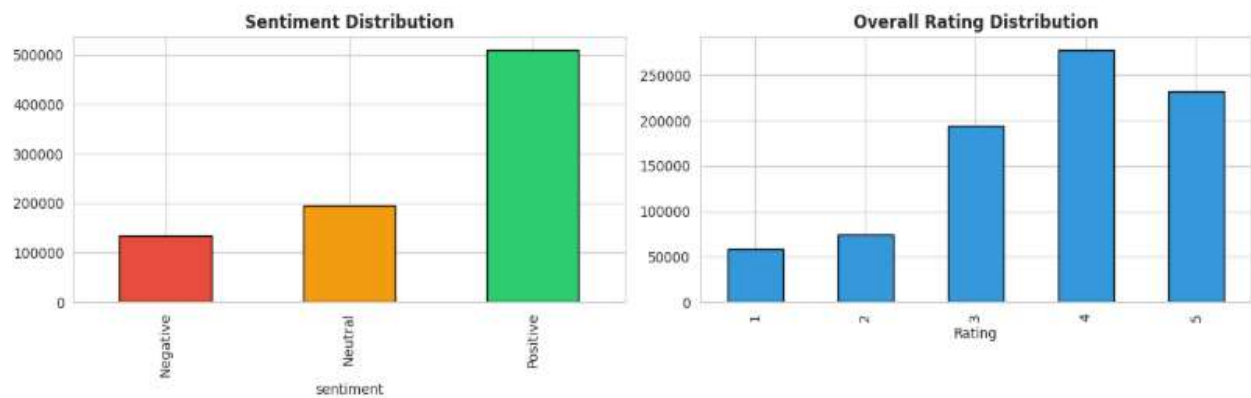


Figure 3 Distribution of sentiment labels and ratings

3.3.3 Bivariate Analysis

Bivariate analysis explored the relationship between individual features and the outcome. Figure 4 relates review length and sentiment composition to the star rating. It shows that review lengths vary across all star ratings, with a few very long reviews appearing as outliers, and it also shows a clear relationship between ratings and sentiment, where low ratings are mostly negative, three star ratings are neutral and high ratings are mostly positive. This confirms that the star rating is a reasonable basis from which to derive a sentiment label.

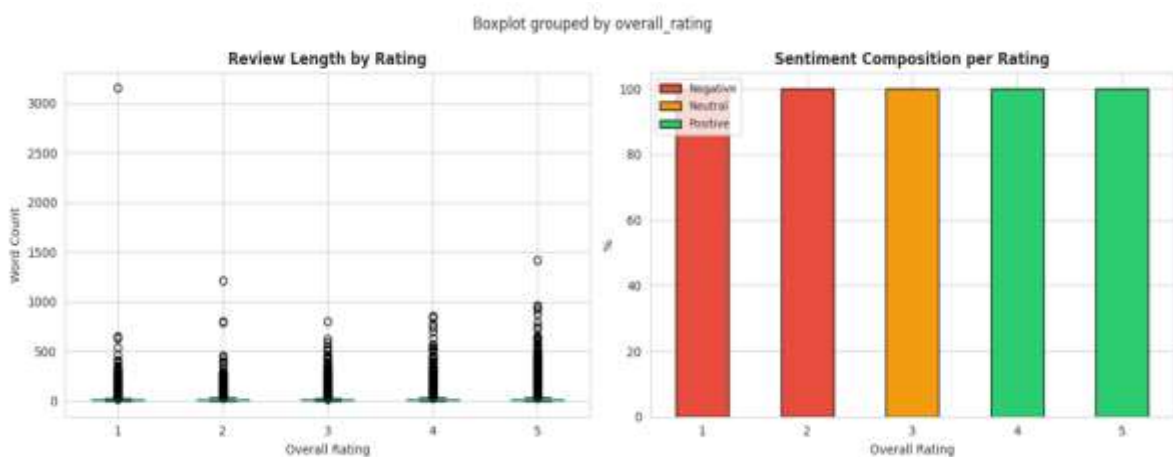


Figure 4 Review length and sentiment distribution by rating

Figure 5 turns to the attrition data and shows that employees who work overtime leave at a noticeably higher rate than those who do not. Roughly a third of the employees working

overtime left, compared with a small fraction of those who did not, which suggests that overtime is an important factor associated with employee turnover.

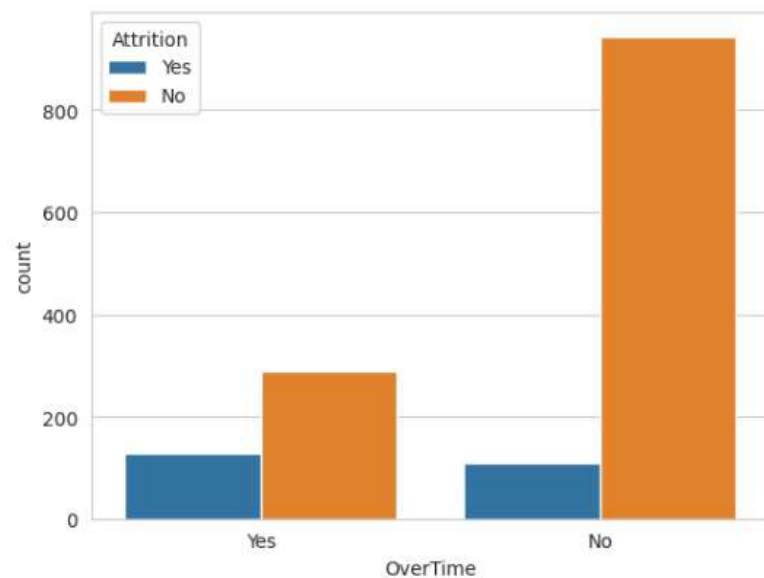


Figure 5 Attrition across overtime status

Figure 6 breaks down attrition by job role. Most employees remained with the company across all roles. However, Sales Representatives, Laboratory Technicians, Sales Executives, and Research Scientists had higher attrition rates. In contrast, managerial and director level roles experienced lower turnover. This pattern suggests that junior and customer facing roles have more retention risk than senior roles.

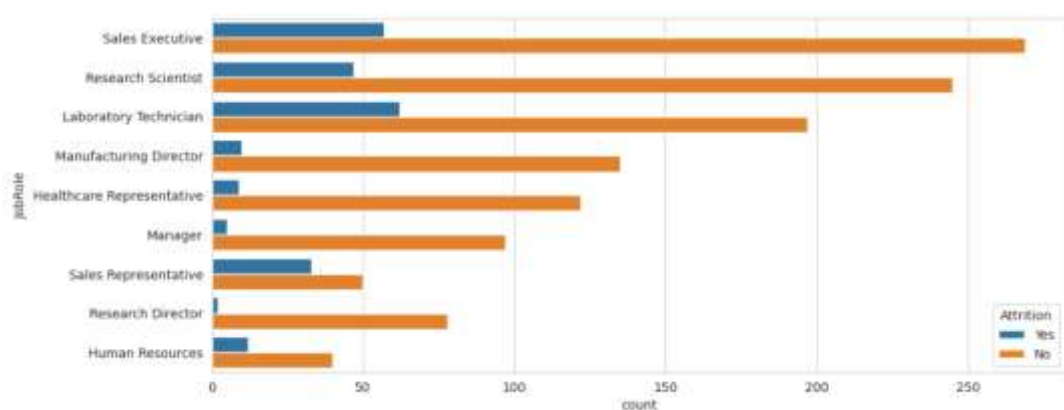


Figure 6 Attrition by employee role

Figure 7 relates business travel to attrition. Employees who travel rarely form the largest group and therefore contribute the highest raw number of attrition cases. Employees who travel

frequently show a higher proportion of attrition relative to their group size, which suggests that extensive travel may contribute to turnover, while employees who do not travel experienced the lowest attrition of all.

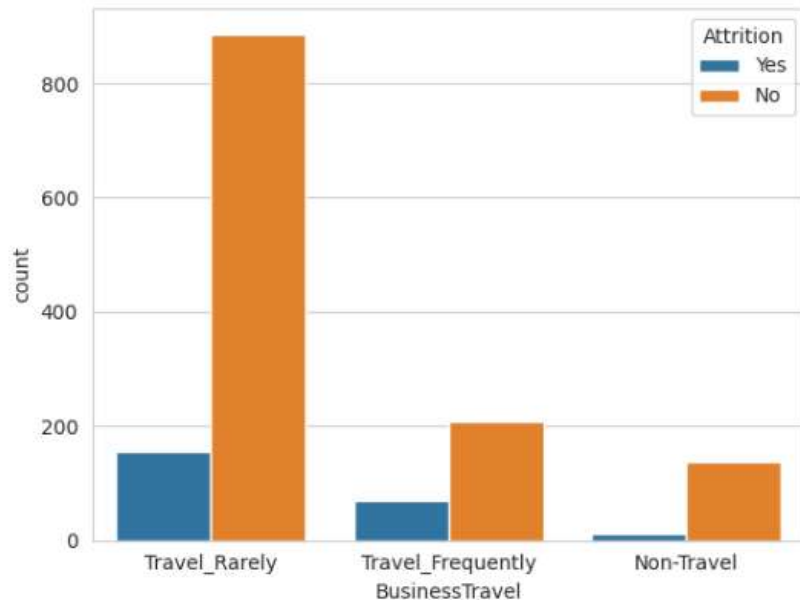


Figure 7 Attrition counts grouped by business travel frequency

Figure 8 presents word clouds of the most frequent terms in positive, neutral and negative reviews. Positive reviews emphasise words such as great, benefit and company, while negative and neutral reviews frequently include Good, People, work, bad and job. Taken together, the reviews indicate that the working environment, benefits and job responsibilities are the main themes influencing employee satisfaction, and they give an early indication of the vocabulary the sentiment models must learn.



Figure 8 Word clouds for each sentiment

3.3.4 Multivariate Analysis

Multivariate analysis was performed in order to examine the relationships among several variables at the same time. Figure 9 presents the correlation heatmap of the numerical features and shows strong positive correlations between JobLevel, MonthlyIncome and TotalWorkingYears. Monthly income was also moderately correlated with YearsAtCompany and Age, which indicates that employee experience and seniority are associated with higher salaries. Most other variables showed weak correlations, suggesting limited linear relationships across the feature set.

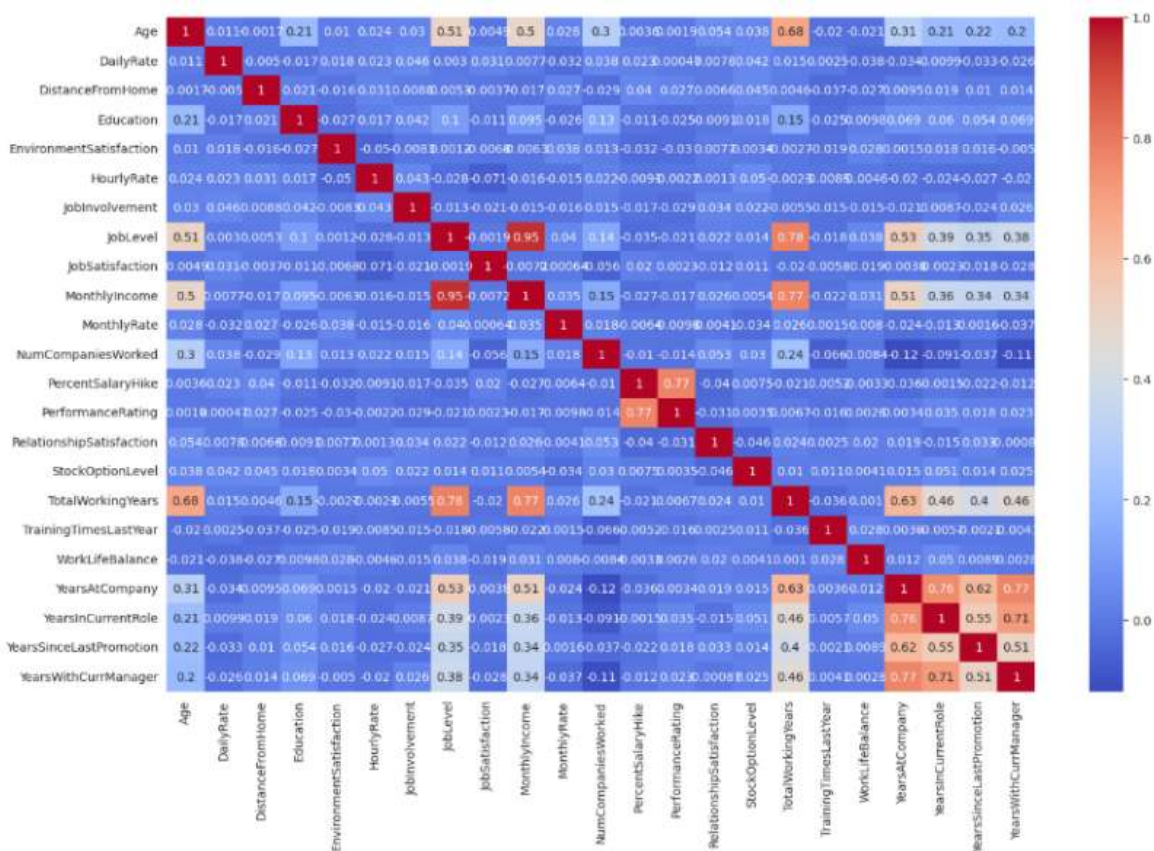


Figure 9 Correlation heatmap of the numeric features in the IBM HR Analytics dataset

Figure 10 displays pairwise relationships among selected key features, coloured by attrition status. The plot shows that employees with lower Age, MonthlyIncome, fewer YearsAtCompany and lower JobSatisfaction are more likely to leave the company, although there is considerable overlap between the two attrition groups. Overall, the multivariate analysis indicates that salary, job level, and work experience are the strongest factors linked to employee characteristics. These factors may also have an indirect impact on attrition.

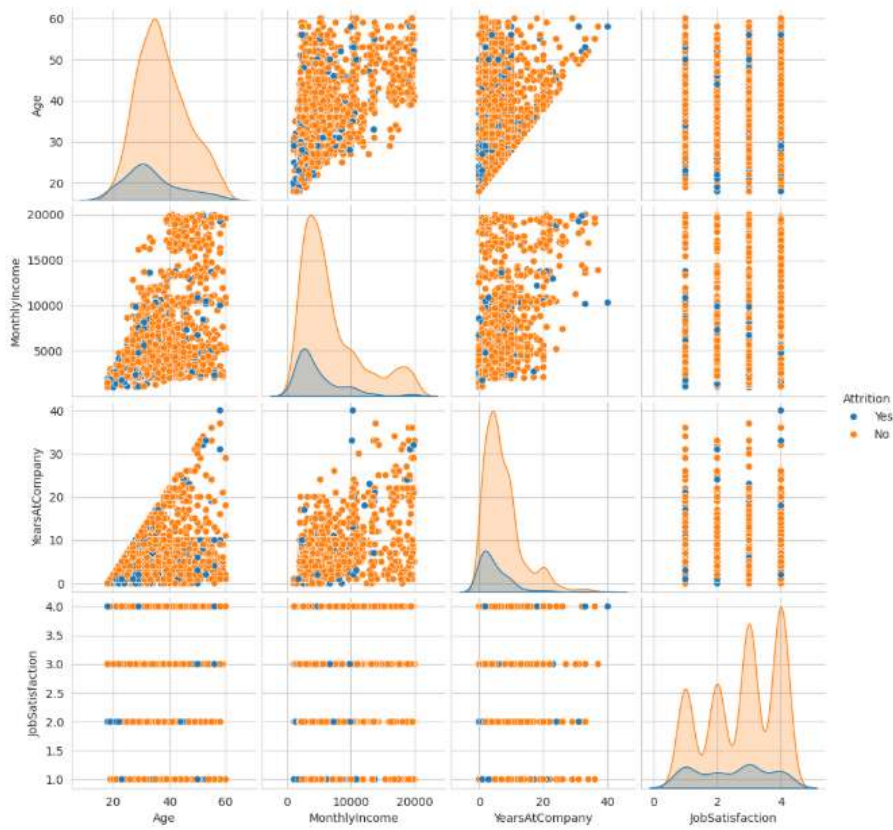


Figure 10 Pairwise relationships between selected key features

3.4 Statistical Testing and Feature Significance

A t-test was performed for each numerical feature and a chi square test was performed for each categorical feature to determine what features most strongly correlate with employee attrition, setting a significance level of $p < 0.05$. The results are given in Table 3 where the p value, decision for test, and interpretation for each feature is reported. The 30 features were tested, and 21 of them were determined to be statistically significant and 9 were determined to be not statistically significant. The results of this investigation provide an objective and evidence-based basis for the selection of features, as opposed to the random selection of features.

Table 3 Feature significance tests for attrition using t-tests and chi-square tests

Feature	p value	Decision	Interpretation
Age	< 0.0001	Reject H0	Significant
BusinessTravel	< 0.0001	Reject H0	Significant
DailyRate	0.0299	Reject H0	Significant

Feature	p value	Decision	Interpretation
Department	0.0045	Reject H0	Significant
DistanceFromHome	0.0028	Reject H0	Significant
Education	0.2293	Fail to reject	Not significant
EducationField	0.0068	Reject H0	Significant
EnvironmentSatisfaction	0.0001	Reject H0	Significant
Gender	0.2906	Fail to reject	Not significant
HourlyRate	0.7931	Fail to reject	Not significant
JobInvolvement	< 0.0001	Reject H0	Significant
JobLevel	< 0.0001	Reject H0	Significant
JobRole	< 0.0001	Reject H0	Significant
JobSatisfaction	0.0001	Reject H0	Significant
MaritalStatus	< 0.0001	Reject H0	Significant
MonthlyIncome	< 0.0001	Reject H0	Significant
MonthlyRate	0.5611	Fail to reject	Not significant
NumCompaniesWorked	0.0955	Fail to reject	Not significant
OverTime	< 0.0001	Reject H0	Significant
PercentSalaryHike	0.6056	Fail to reject	Not significant
PerformanceRating	0.9119	Fail to reject	Not significant
RelationshipSatisfaction	0.0787	Fail to reject	Not significant
StockOptionLevel	< 0.0001	Reject H0	Significant
TotalWorkingYears	< 0.0001	Reject H0	Significant
TrainingTimesLastYear	0.0226	Reject H0	Significant
WorkLifeBalance	0.0142	Reject H0	Significant
YearsAtCompany	< 0.0001	Reject H0	Significant
YearsInCurrentRole	< 0.0001	Reject H0	Significant
YearsSinceLastPromotion	0.2058	Fail to reject	Not significant
YearsWithCurrManager	< 0.0001	Reject H0	Significant

To examine the effect of statistical feature selection on predictive performance, each deep learning model described later in Section 3.6 is trained under two experimental settings. In the first setting the models are trained using all 30 features. In the second setting the models are trained using only the 21 statistically significant features identified through the hypothesis

testing process reported in Table 3. Comparing these two approaches makes it possible to evaluate whether excluding the 9 features that were not significant improves prediction accuracy, computational efficiency and the ability of the models to generalise to unseen data.

The same evaluation strategy is followed during cross dataset validation. Each deep learning model is evaluated twice, once with all 30 features and once with only the 21 statistically significant features. Applying the same validation procedure to both feature sets ensures a fair and consistent comparison, which allows the impact of statistical feature selection on model performance to be assessed reliably.

3.5 Data Preprocessing and Feature Engineering

Once the datasets had been understood, both were prepared for modelling. Preprocessing was carried out separately for the structured and textual data, but both followed the same underlying principle, which is that any transformation capable of leaking information from the validation or test set into training was deliberately avoided.

3.5.1 Preprocessing the HR Attrition Data

The HR dataset contained no missing values, outliers and duplicate records, so no imputation or removal of duplicates was required. Four constant or purely identifying columns, namely EmployeeCount, EmployeeNumber, Over18 and StandardHours, were removed because they carry no discriminative information. The target was encoded numerically, with 1 representing a leaver and 0 a stayer, and the remaining categorical columns were converted to numeric form using label encoding. A configurable feature switch was built into the pipeline so that the models could be trained either on the complete set of 30 features or on the reduced subset of 21 statistically significant features, which allows the effect of feature selection to be measured directly.

The data was divided into training, validation and test sets in an 80/10/10 ratio using stratified splitting in order to preserve the class balance, and all numeric features were standardised with a scaler fitted only on the training data. To handle the class imbalance, the complete feature variant deliberately trained on the raw distribution and compensated at the evaluation stage by relying on Precision, Recall, F1 score and ROC-AUC rather than Accuracy alone. Finally, the scaled arrays were converted into PyTorch tensors and wrapped in DataLoader objects for mini batch training.

3.5.2 Preprocessing the Glassdoor Review Text

The raw Glassdoor data did not contain explicit sentiment labels, so these were derived from the associated star ratings. Reviews with ratings of four or five were labelled as positive, while ratings of one or two were labelled as negative. Reviews with a rating of three were treated as neutral and excluded, which allowed the task to be formulated as a balanced binary classification problem. To address the original class imbalance, an equal number of positive and negative reviews were randomly sampled.

Each review was then preprocessed by converting the text to lowercase and removing URLs, emojis, punctuation and common stop words. The remaining words were lemmatised to their base forms before being tokenised using a vocabulary of 20,000 words. The resulting sequences were padded or truncated to a fixed length in order to enable efficient batch processing during model training.

The processed reviews were subsequently divided into training, validation and test sets using a stratified split, which preserves the class distribution across each subset. No hand crafted linguistic features were engineered for the text models. Instead, feature representations were learned automatically through the embedding, convolutional and recurrent layers, which enables the models to capture semantic and contextual information directly from the review text.

3.6 Model Implementation

The system was implemented as two parallel pipelines that share the same five stage workflow. Each model was first initialised together with its loss function, optimiser and learning rate scheduler. It was then trained using an early stopping strategy in order to prevent overfitting, and evaluated separately on the training, validation and held out test sets. Each model was afterwards explained using one or more explainability techniques, and finally saved to disk together with its preprocessing objects for deployment.

3.6.1 Attrition Models Based on the FT-Transformer and the Tabular ResNet

Two models were implemented for attrition prediction, both trained on the identical preprocessed and scaled feature set so that any difference in their results could be attributed to the architecture rather than the data. The FT-Transformer converts each structured feature into a small learned vector, allows these vectors to exchange information through several transformer encoder blocks, and combines them using an attention pooling layer before making

a prediction. The Tabular ResNet instead passes the features through a series of residual blocks, each adding its output back onto its input, which allows a deeper network to be trained without the gradient vanishing. Table 4 summarises the FT-Transformer architecture together with the parameter count of each component, and Table 5 summarises the Tabular ResNet in the same way.

Table 4 FT-Transformer architecture summary

Component	Output shape	Parameters
Feature tokeniser that maps each of 30 features to a 64-dimensional token	(30, 64)	3,840
CLS token, a learnable classification token	(1, 64)	64
Four transformer encoder blocks with 8 head attention, layer normalisation and a GELU feed forward network	(31, 64) per block	199,936
Attention pooling layer that combines 31 tokens into one vector	(64)	65
Classifier head with three linear layers, batch normalisation, GELU and dropout	(1)	58,753
Total trainable parameters	--	262,658

As Table 4 shows, most of the capacity of the FT-Transformer sits in its four encoder blocks, which hold roughly 200,000 of the 262,658 trainable parameters. The tokeniser and the attention pooling layer are comparatively small, which means the model spends its capacity on learning interactions between features rather than on encoding the features themselves.

Table 5 Tabular ResNet architecture summary

Component	Description
Input projection	A linear layer that maps the 30 features into a 128-dimensional representation
Four residual blocks	Batch normalisation, linear expansion to 256 units, GELU, dropout, linear contraction to 128 units, dropout and a skip connection
Output head	Batch normalisation, GELU activation and a final linear layer producing one attrition logit
Total trainable parameters	269,057

Table 5 shows that the Tabular ResNet reaches a comparable total of 269,057 trainable parameters through a very different route. Its depth comes from four residual blocks that repeatedly expand and contract the representation, so the two models are close in size and the comparison between them reflects architecture rather than capacity.

Both models were trained with binary cross entropy loss and the AdamW optimiser at a learning rate of 0.001 with a small weight decay term, using a step scheduler that halved the learning rate every 20 epochs. Training ran for up to 100 epochs, but early stopping monitored validation accuracy and halted training once 10 consecutive epochs passed without meaningful improvement, restoring the best weights. The full training configuration is given in Table 6.

Table 6 Training configuration for the two attrition models

Setting	FT-Transformer	Tabular ResNet
Loss function	Binary cross entropy with logits	Binary cross entropy with logits
Optimiser	AdamW with lr = 0.001 and weight decay = 1e-5	AdamW with lr = 0.001 and weight decay = 1e-5
Learning rate scheduler	Step decay, halved every 20 epochs	Step decay, halved every 20 epochs
Maximum epochs	100	100
Early stopping	Patience of 10 on validation accuracy	Patience of 10 on validation accuracy
Batch delivery	Mini batch via PyTorch DataLoader	Mini batch via PyTorch DataLoader

Table 6 confirms that both attrition models were trained under identical conditions, which keeps the later comparison fair.

3.6.2 Sentiment Models Based on the BiLSTM and the CNN-BiGRU with Attention

Two models were implemented for sentiment analysis, both trained on the identical balanced and tokenised review data. The BiLSTM reads each review both forwards and backwards through two stacked bidirectional LSTM layers before reducing the sequence to a single vector using global max pooling. The CNN-BiGRU model begins with two convolutional layers that extract short local word patterns, passes these through two stacked bidirectional GRU layers in order to capture sequence context, and finally through a custom additive attention layer that focuses on the words most relevant to the sentiment decision. Table 7 summarises the BiLSTM architecture and Table 8 summarises the CNN-BiGRU model with attention.

Table 7 BiLSTM architecture summary

Layer	Output shape	Parameters
Embedding	(80, 128)	2,560,000
Spatial dropout	(80, 128)	0
Bidirectional LSTM with 64 units	(80, 128)	98,816

Layer	Output shape	Parameters
Bidirectional LSTM with 32 units	(80, 64)	41,216
Global max pooling	(64)	0
Dense with 64 units and ReLU	(64)	4,160
Batch normalisation	(64)	256
Dropout	(64)	0
Dense with 2 units and softmax	(2)	130
Total and trainable parameters	--	2,704,578 / 2,704,450

Table 8 CNN-BiGRU with attention architecture summary

Layer	Output shape	Parameters
Input	(80)	0
Embedding	(80, 128)	2,560,000
Spatial dropout	(80, 128)	0
Convolutional layer with 128 filters	(80, 128)	49,280
Convolutional layer with 64 filters	(80, 64)	24,640
Bidirectional GRU with 64 units	(80, 128)	49,920
Bidirectional GRU with 32 units	(80, 64)	31,104
Attention pooling, a custom layer	(64)	4,224
Dense with 64 units and ReLU	(64)	4,160
Batch normalisation	(64)	256
Dense with 2 units and softmax	(2)	130
Total and trainable parameters	--	2,723,714 / 2,723,586

Comparing Table 7 and Table 8 shows that both models share the same embedding dimension and a fixed input length of eighty tokens, so the comparison between them remains fair. In each case the embedding layer accounts for the overwhelming majority of the parameters. The CNN-BiGRU model carries slightly more parameters in total, 2,723,586 against 2,704,450, because of its convolutional and attention layers, and this modest difference is what buys it the ability to weight individual words.

Both text models were trained with sparse categorical cross entropy loss and the Adam optimiser at a learning rate of 0.001, using two callbacks. An early stopping callback restored the best weights after three epochs without improvement, and a learning rate reduction callback halved the learning rate when validation loss plateaued. Balanced class weights were applied as an additional safeguard against residual imbalance. Table 9 gives the full configuration.

Table 9 Training configuration for the two sentiment models

Setting	BiLSTM	CNN-BiGRU with attention
Loss function	Sparse categorical cross entropy	Sparse categorical cross entropy
Optimiser	Adam with lr = 0.001	Adam with lr = 0.001
Callbacks	Early stopping with patience 3 and ReduceLROnPlateau with factor 0.5 and patience 1	Early stopping with patience 3 and ReduceLROnPlateau with factor 0.5 and patience 1
Maximum epochs	12	12
Batch size	256	256
Class weighting	Balanced class weights applied	Balanced class weights applied

As Table 9 records, both sentiment models were trained under the same settings, so the difference in their results can be attributed to architecture alone.

3.7 Explainability

To make the attrition models transparent, several complementary techniques were implemented, taking account of the architectural differences between the two models. For the FT-Transformer the attention mechanism was examined by visualising the attention weights assigned to each feature, which revealed both the features the model focused on for individual employees and their average importance across the test set. Because the Tabular ResNet has no attention mechanism, permutation feature importance was used instead, measuring the drop in accuracy when each feature was randomly shuffled. To provide a model agnostic explanation applicable to both models, SHAP with the Kernel Explainer was implemented, quantifying the contribution of each feature to individual predictions. A custom real time explanation function was also developed for the FT-Transformer, classifying the risk level of an employee and summarising the features that contributed most to the prediction in human readable form.

For the sentiment models, LIME was applied to both the BiLSTM and the CNN-BiGRU model in order to identify the specific words within a single review that most influenced its

classification, while SHAP was applied to the BiLSTM to provide a global view of the most influential words across a sample of reviews. A custom real time explanation function was developed for the BiLSTM to analyse newly submitted feedback, predict its sentiment and associated risk level, highlight the influential words and recommend an appropriate HR intervention. Together these techniques provide both local and global interpretability, increasing user confidence and supporting the practical adoption of explainable artificial intelligence in workforce analytics.

3.8 Deployment Artefacts

After training, evaluation and explanation, both pipelines were prepared for deployment by saving the trained models together with all necessary preprocessing components, so that the models can be reused for inference without retraining. For the attrition pipeline the saved artefacts included the FT-Transformer and Tabular ResNet checkpoints, the feature scaler, the categorical encoder, the feature name list and the list of statistically significant features. For the sentiment pipeline the deployment package contained the BiLSTM and CNN-BiGRU models, the text tokeniser, the sequence length configuration, the vocabulary settings and the sentiment label mapping. Preserving both the models and their preprocessing objects allows the complete system to be deployed efficiently as an interactive Hugging Face Spaces application while maintaining consistency between the training and inference stages.

3.9 Summary

This chapter has described the end to end implementation of both pipelines, from data understanding and exploratory analysis through preprocessing, statistical feature selection and the design and training of four deep learning models, to explainability and the preparation of deployment artefacts. Although the two pipelines process different data modalities and use different frameworks, they follow a consistent methodology and converge on a shared evaluation, explanation and deployment stage. The next chapter evaluates the resulting models by comparing their performance using the complete feature set and the statistically significant feature set, together with cross dataset validation results, and relates the findings to the existing literature.

Chapter 4 Evaluation and Results

This chapter evaluates the four deep learning models and weighs their strengths and weaknesses. Each model was assessed on the training, validation and held out test sets, and the models within each pipeline were compared using Accuracy, Precision, Recall, F1 score and ROC-AUC, supported by confusion matrices and learning curves. Because the attrition target is imbalanced, particular attention is paid to performance on the minority Leave class rather than to overall accuracy alone. The results are reported first for the complete feature set, then for the statistically significant feature subset, and finally for a second HR dataset used for cross dataset validation, before the deployed application is presented and the outcomes are related to existing work.

4.1 Attrition Results with the Complete Feature Set

Both FT-Transformer and Tabular ResNet were initially trained with all the 30 features. The training and validation curves for the FT-Transformer are presented in figure 11. The two loss curves converge and the two accuracy curves increase together but then level off, supporting the assertion that there was no serious overfitting and that training was stopped at an appropriate point via early stopping.



Figure 11 FT-Transformer training and validation loss and accuracy

Figure 12 shows the confusion matrix of the same model and highlights the impact of the dataset's class imbalance. The correct classification of most employees who remained is achieved while a significant proportion of those who departed are correctly predicted as stayers, suggesting poorer performance on the minority group.

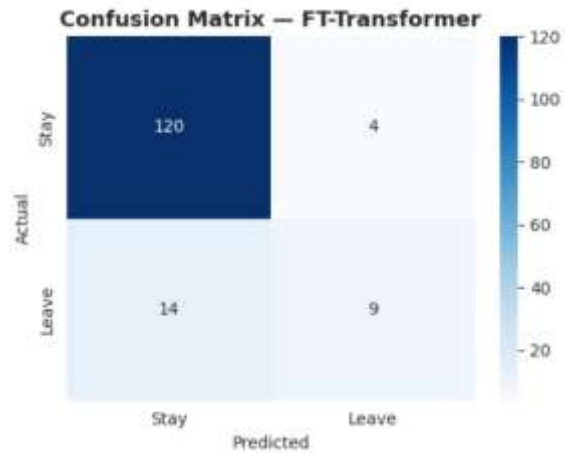


Figure 12 FT-Transformer test set confusion matrix using all features

Table 10 reports the full classification results behind that matrix. The model reached a test accuracy of 87.76%, and Validation accuracy of the 90.48, although it identified the majority Stay class very reliably with a recall of 0.97, its recall on the minority Leave class remained limited at 0.39. The gap between the two classes reflects the difficulty imposed by the class imbalance rather than a failure of the architecture.

Table 10 Evaluation results of the FT-Transformer model with the complete feature set

Class or Metric	Precision or Value	Recall	F1 Score	Support
Training accuracy	92.18%	--	--	--
Validation accuracy	90.48%	--	--	--
Test accuracy	87.76%	--	--	--
Stay	0.90	0.97	0.93	124
Leave	0.69	0.39	0.50	23
Macro average	0.79	0.68	0.72	147
Weighted average	0.86	0.88	0.86	147

Figure 13 shows the Tabular ResNet training curves, which are noticeably noisier than those of the FT-Transformer and take longer to settle.

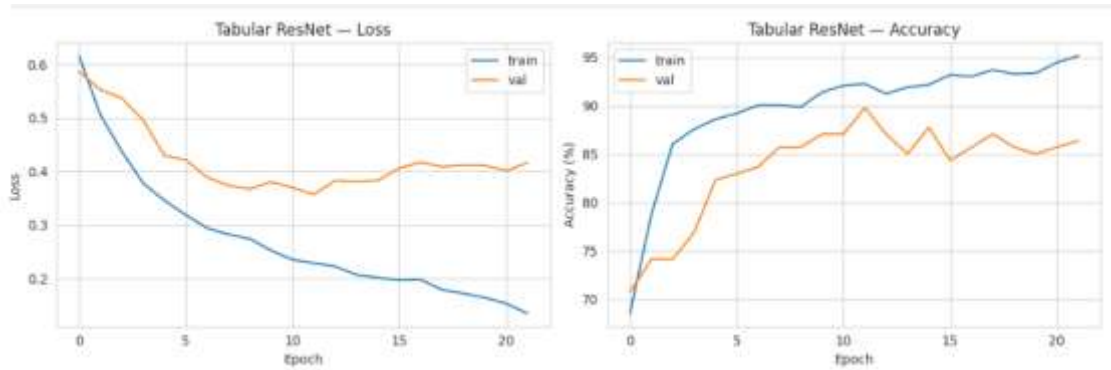


Figure 13 Tabular ResNet training and validation loss and accuracy using all features

Figure 14 shows the confusion matrix of the Tabular ResNet. The same pattern appears as before, with an even greater number of employees who left being misclassified as stayers. This indicates that the model is more strongly influenced by the class imbalance in the dataset, resulting in reduced performance on the minority class despite good overall classification of the majority class.

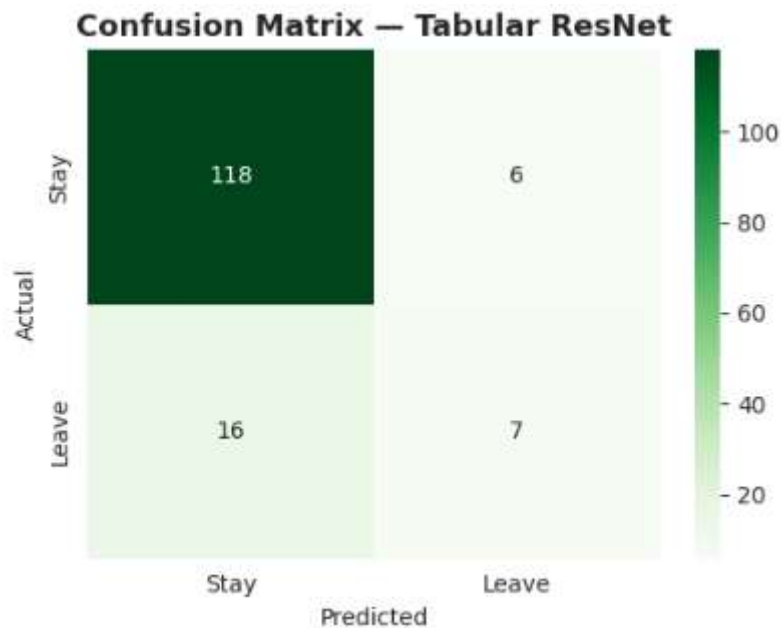


Figure 14 Tabular ResNet test set confusion matrix using all features

As Table 11 reports, the Tabular ResNet reached a lower test accuracy of 85.03% with a ROC-AUC of 0.740 and weaker minority class recall of 0.30. This confirms that the attention based FT-Transformer captured the feature interactions in this dataset more effectively than the residual architecture.

Table 11 Tabular ResNet evaluation results using all features

Class or Metric	Precision or Value	Recall	F1 Score	Support
Test accuracy	85.03%	--	--	--
ROC-AUC	0.740	--	--	--
Stay	0.88	0.95	0.91	124
Leave	0.54	0.30	0.39	23
Macro average	0.71	0.63	0.65	147
Weighted average	0.83	0.85	0.83	147

Figure 15 places the two models side by side across every metric, which makes the advantage of the FT-Transformer on accuracy, precision and F1 score easy to see at a glance.

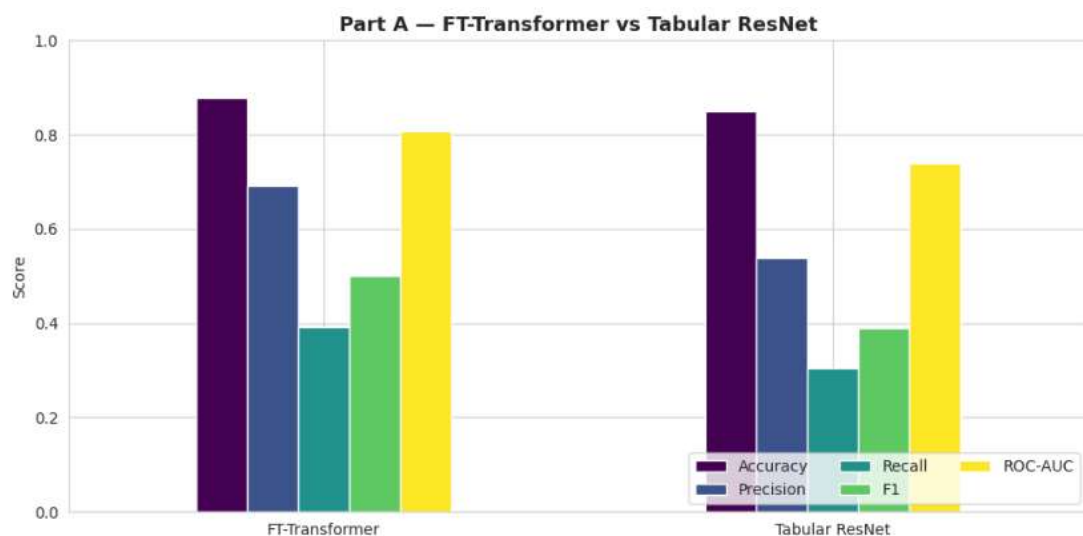


Figure 15 Comparison of FT-Transformer and Tabular ResNet performance using all features

Figure 16 compares two different views of feature importance, the attention weights taken from the FT-Transformer and the permutation importance measured for the Tabular ResNet. The two rankings overlap only partly. OverTime is prominent in both, but the Tabular ResNet leans on YearsSinceLastPromotion and Age, while the FT-Transformer distributes attention towards Department, MonthlyIncome and EnvironmentSatisfaction. Such a difference is to be expected since the permutation importance is the decrease in accuracy when a feature is scrambled, while the attention is the extent to which a feature is attended to, and a feature may be attended to without being decisive.

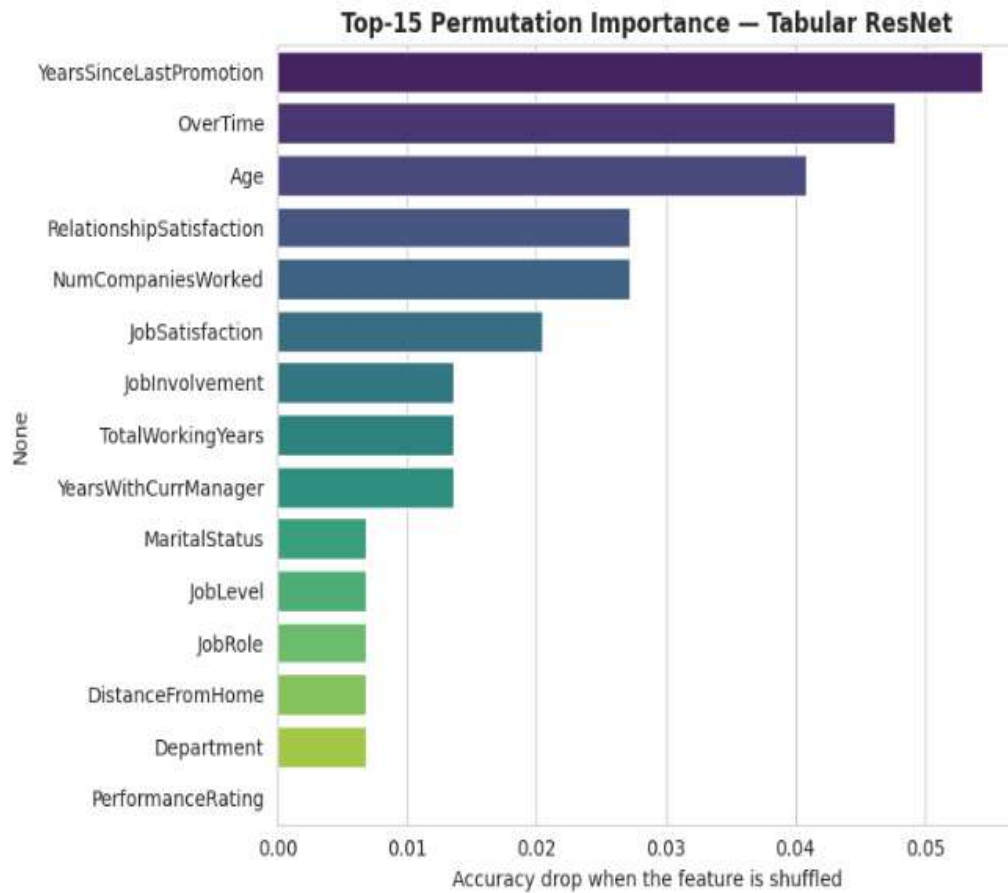


Figure 16 Feature importance comparison between FT-Transformer attention weights and Tabular ResNet permutation importance

Figure 17 shows the average attention that the FT-Transformer paid to each feature across the whole test set. Department, MonthlyIncome, EnvironmentSatisfaction and OverTime receive the most weight, while MonthlyRate and YearsAtCompany receive the least. The spread is narrow, running roughly from 0.02 to 0.06, which suggests that the model draws on many features rather than relying on a few dominant ones.

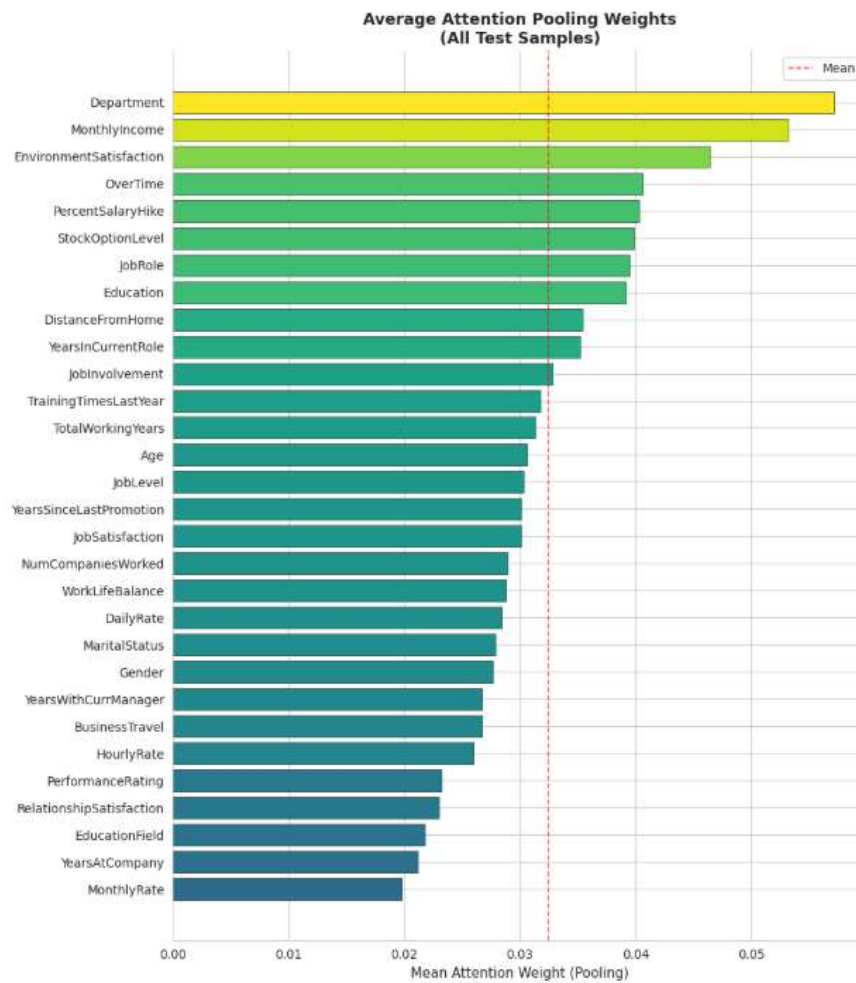


Figure 17 Average attention pooling weights assigned by the FT-Transformer across all test employees

Figure 18 shows how each feature influences the model's attrition predictions, including both the strength and direction of its impact. Positive SHAP values indicate that a feature contributes towards predicting that an employee will leave, while the colour represents the value of that feature. OverTime has the strongest influence among the features shown, with employees working overtime having a much stronger shift towards attrition predictions compared with those not working overtime. Other factors, such as lower stock option levels, longer commute distances, younger age, lower job levels and lower environment satisfaction, also contribute towards attrition predictions, which agrees with patterns commonly reported in employee turnover studies. However, the direction of categorical features that were converted into numerical values, such as Department and JobRole, should not be interpreted directly, as the colour order represents the encoding scheme rather than an actual numerical scale. For clarity, only the twenty most important features are included in the plot.

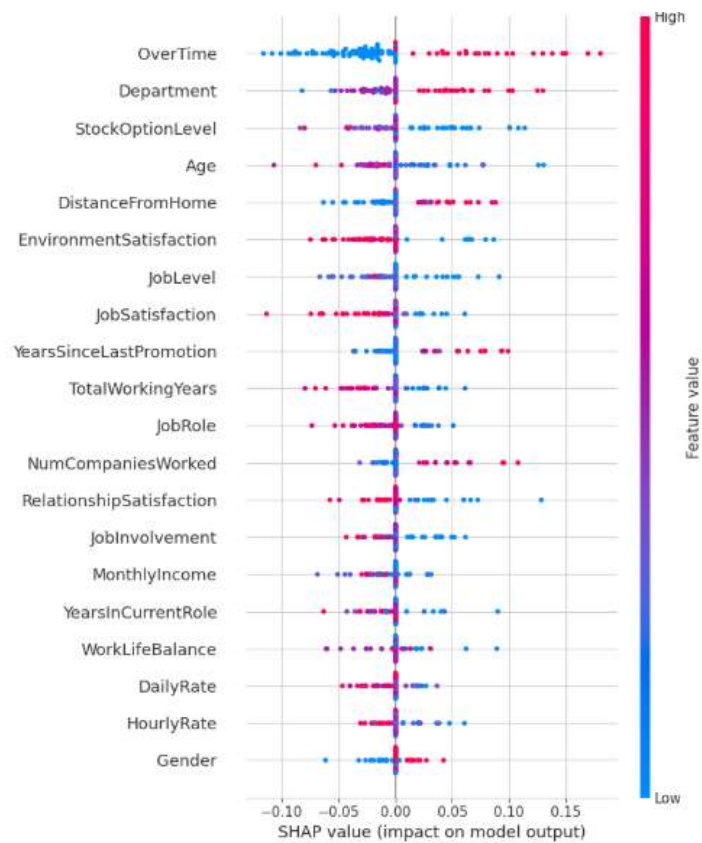


Figure 18 SHAP summary plot for the FT-Transformer

Figure 19 and Figure 20 split the attention weights by predicted outcome. Figure 19 covers employees predicted to stay, where attention concentrates on Department, MonthlyIncome and EnvironmentSatisfaction and Overtime.

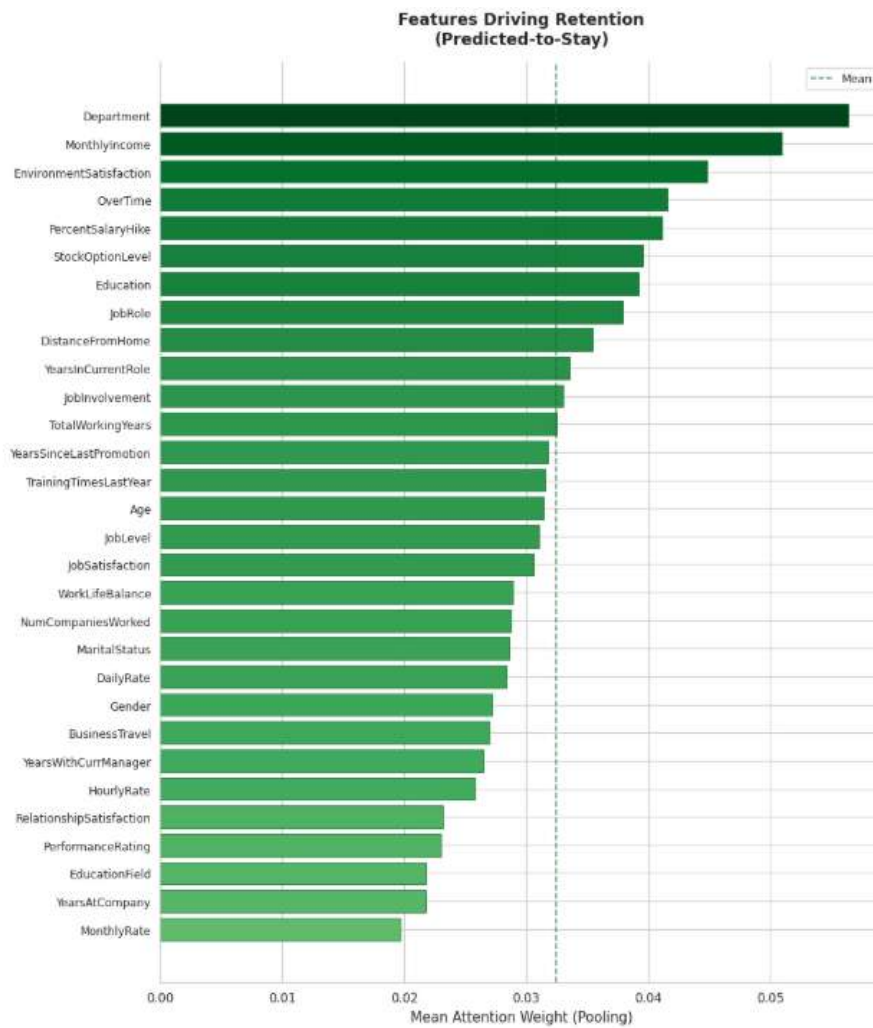


Figure 19 Features driving predictions towards employee retention

Figure 20 covers employees predicted to leave. Here MonthlyIncome moves to the front, followed by Department and EnvironmentSatisfaction, with YearsInCurrentRole and JobRole rising sharply. The same features dominate both outcomes, but their ordering shifts with the prediction, which suggests the model reweights rather than replaces the evidence it uses.

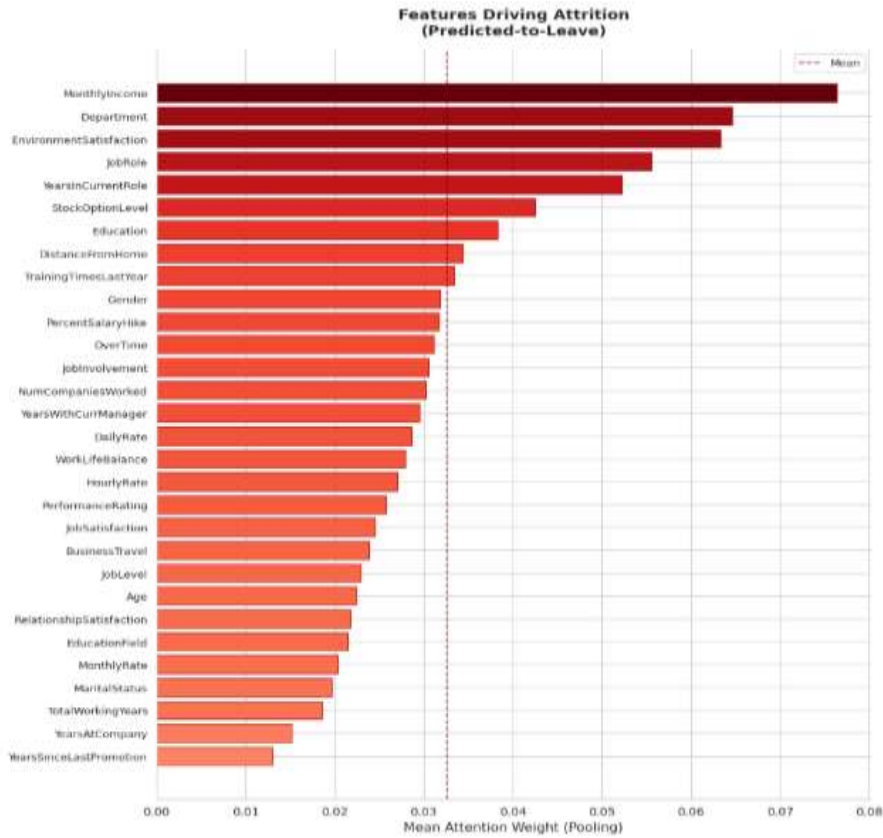


Figure 20 Features most strongly driving predictions towards attrition, that is employees predicted to leave

4.2 Sentiment Results with the Complete Dataset

The BiLSTM and the CNN-BiGRU models were trained on the balanced set of employee reviews. Figure 21 shows the BiLSTM training curves, which climb steadily before levelling off, with validation accuracy tracking training accuracy closely across the small number of epochs used.



Figure 21 BiLSTM training and validation loss and accuracy curves

Figure 22 shows the confusion matrix of the BiLSTM. Unlike the attrition task, the errors are shared almost evenly between the positive and negative classes, which reflects the balanced review data and shows that neither sentiment is systematically favoured.

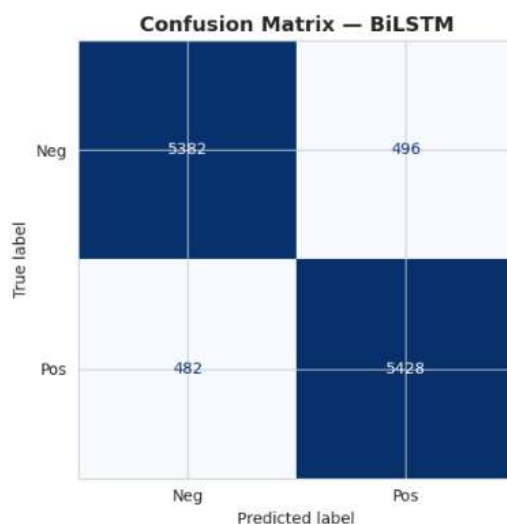


Figure 22 BiLSTM sentiment model test set confusion matrix

As Table 12 reports, the BiLSTM reached a test accuracy of 91.70% with balanced precision and recall across both classes. This is a much stronger result than on the attrition task because the review dataset is both large and balanced.

Table 12 BiLSTM sentiment model evaluation results on the complete dataset

Class or Metric	Precision or Value	Recall	F1 Score	Support
Training accuracy	93.44%	--	--	--
Validation accuracy	92.08%	--	--	--
Test accuracy	91.70%	--	--	--
Negative	0.918	0.916	0.917	5,878
Positive	0.916	0.918	0.917	5,910
Macro average	0.917	0.917	0.917	11,788

Figure 23 shows the word level SHAP explanation for the BiLSTM. Words such as lack, stressful and bureaucratic carry the largest negative weight and push a review towards the

negative class, whereas stable is the clearest positive signal. The explanation confirms that the model has learned workplace specific vocabulary rather than generic sentiment words.

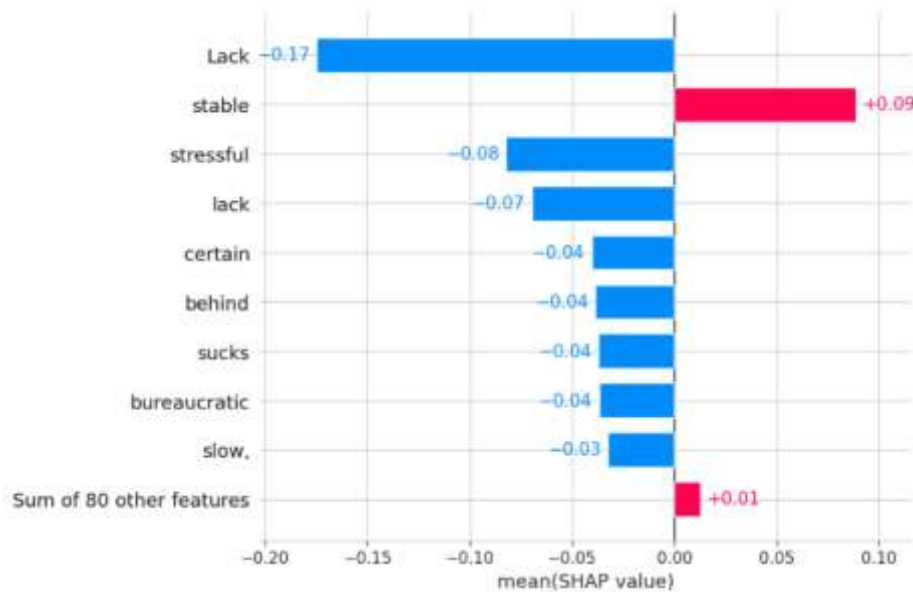


Figure 23 Word-level SHAP explanation for the BiLSTM sentiment model

Figure 24 shows the CNN-BiGRU training curves, which converge faster than those of the BiLSTM.

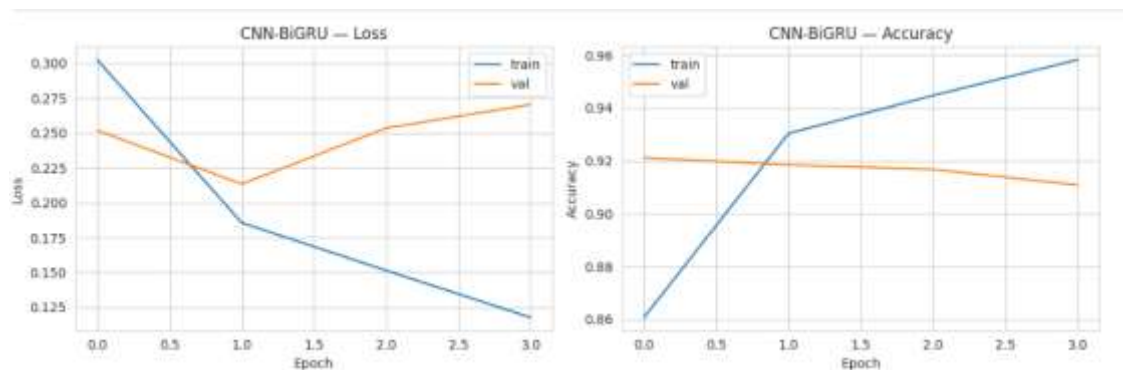


Figure 24 CNN-BiGRU with attention training and validation loss and accuracy curves

Figure 25 shows the confusion matrix of the CNN-BiGRU model, which has the strongest diagonal of any model in the project, with only a small number of errors on either side of it.

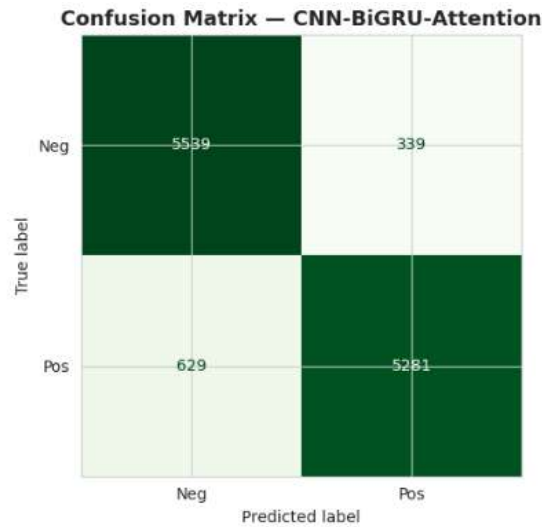


Figure 25 CNN-BiGRU with attention test set confusion matrix

As Table 13 reports, the CNN-BiGRU model with attention achieved the best sentiment result, with a test accuracy of 91.79% and a ROC-AUC of 0.972, slightly ahead of the BiLSTM. The margin is small, but the model also recovers more of the negative reviews, with a recall of 0.942 on that class.

Table 13 CNN-BiGRU with attention model evaluation results on the complete dataset

Class or Metric	Precision or Value	Recall	F1 Score	Support
Test accuracy	91.79%	--	--	--
ROC-AUC	0.972	--	--	--
Negative	0.898	0.942	0.920	5,878
Positive	0.940	0.894	0.916	5,910
Macro average	0.919	0.918	0.918	11,788

Figure 26 compares the performance of the BiLSTM and CNN-BiGRU with Attention models across the five-evaluation metrics. Both models perform well, with scores above 0.90 for most metrics. The CNN-BiGRU with Attention model achieves slightly better Precision and ROC-AUC, while the BiLSTM records a marginally higher Recall. Overall, the results show that the CNN-BiGRU with Attention model delivers the stronger overall performance.

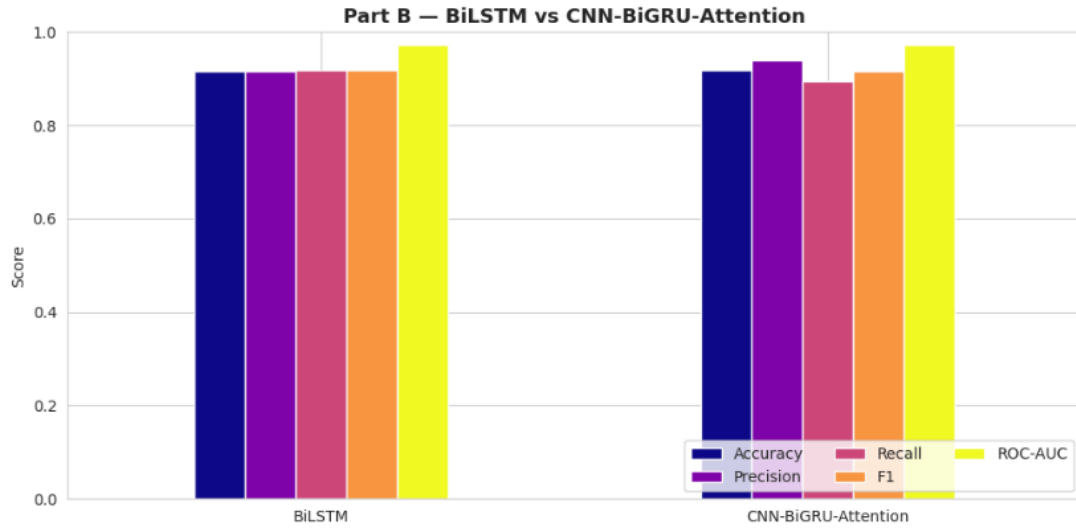


Figure 26 Comparison of BiLSTM and CNN-BiGRU with Attention model performance

4.3 Attrition Results with the Significant Feature Subset

To test whether feature selection improves the attrition models, both were retrained on only the 21 statistically significant features identified in Section 3.4 and listed in Table 3. Figure 27 shows the FT-Transformer training curves for this configuration, which converge slightly earlier than with the complete feature set because the model has fewer inputs to learn from.



Figure 27 FT-Transformer training and validation curves using the significant feature subset

Figure 28 shows the confusion matrix of the reduced model. Slightly fewer stayers are misclassified than with the complete feature set, although leavers remain the harder class to catch.

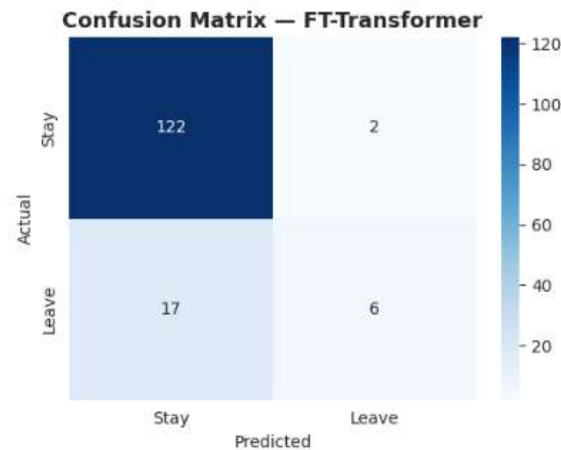


Figure 28 Confusion matrix for the FT-Transformer with the significant feature subset

As Table 14 reports, the FT-Transformer reached a test accuracy of 87.07%, marginally below its complete feature result, but its precision on the Leave class rose from 0.69 to 0.75. Removing uninformative features therefore produced leaver predictions that were more confident when made, though not more numerous, since recall on that class fell from 0.39 to 0.26.

Table 14 Evaluation results of the FT-Transformer model with the significant feature subset

Class or Metric	Precision or Value	Recall	F1 Score	Support
Training accuracy	90.56%	--	--	--
Validation accuracy	89.80%	--	--	--
Test accuracy	87.07%	--	--	--
Stay	0.88	0.98	0.93	124
Leave	0.75	0.26	0.39	23
Macro average	0.81	0.62	0.66	147
Weighted average	0.86	0.87	0.84	147

Figure 29 shows the average attention pooling weights of the significant features in the FT-Transformer model. BusinessTravel, MonthlyIncome, YearsInCurrentRole, JobRole, and OverTime receive the highest attention weights, indicating that they contribute more to the model’s predictions than the other selected features.

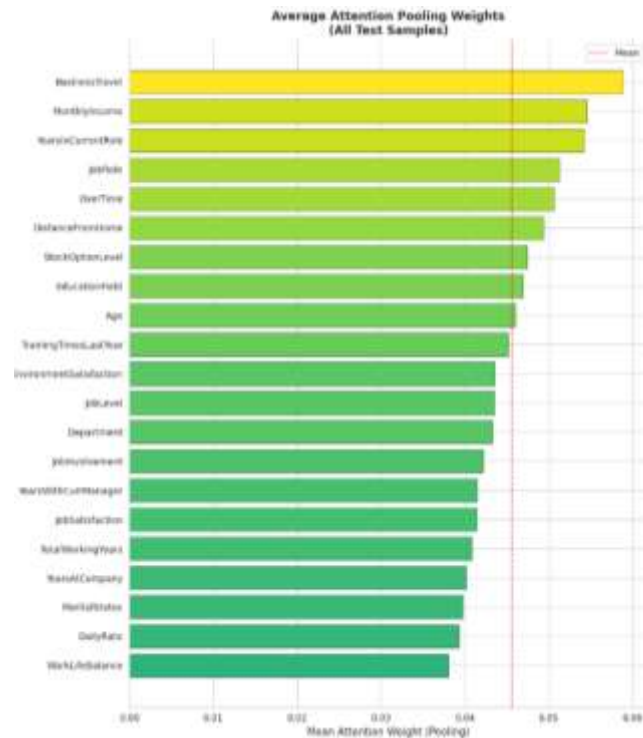


Figure 29 Average Attention Pooling Weights of Significant Features in the FT-Transformer Model

Figure 30 shows how the SHAP values spread across the individual features. The wider the horizontal spread of a feature, the more its influence varies from one employee to another, which is why OverTime and StockOptionLevel stretch furthest.

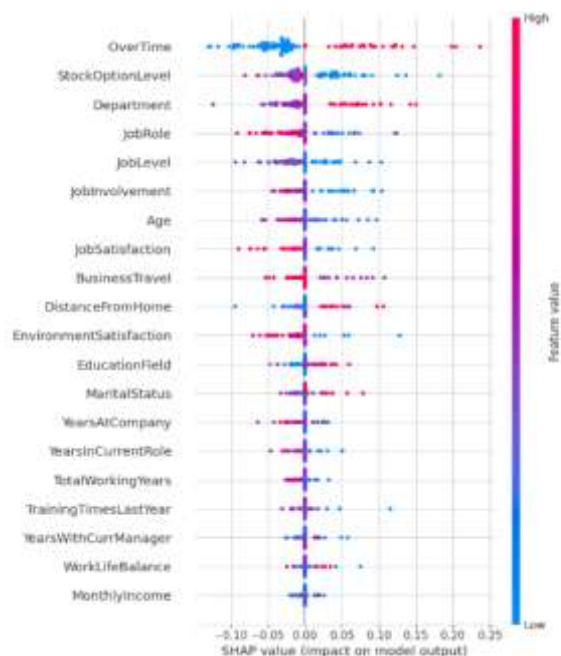


Figure 30 SHAP value distribution across features for the significant feature FT-Transformer

Figure 31 illustrates the most important features used by the FT-Transformer model to predict employee retention and attrition. The left chart represents employees who are likely to stay, while the right chart represents employees who are likely to leave. The dashed line indicates the average attention weight, with higher values showing a stronger influence on the model's predictions. The results show that MonthlyIncome, YearsInCurrentRole, BusinessTravel, DistanceFromHome, and JobRole are among the most influential factors in both predictions, suggesting that salary, career growth, commuting distance, and job-related conditions play a major role in employee retention and turnover.

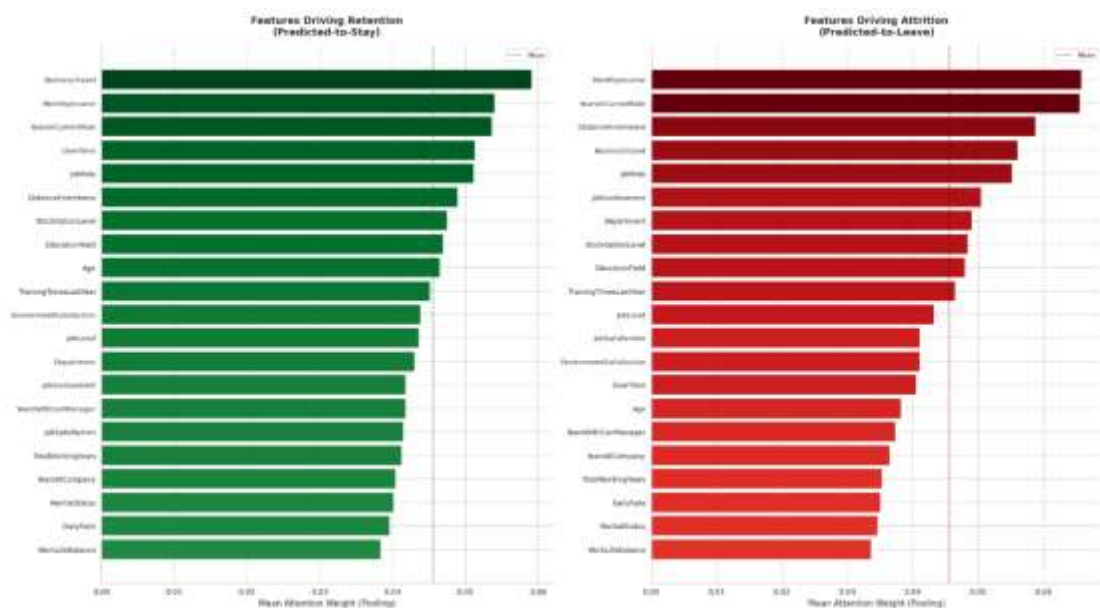


Figure 31 FT-Transformer Attention-Based Feature Importance for Employee Retention and Attrition

Figure 32 shows the Tabular ResNet training curves for the reduced feature set.

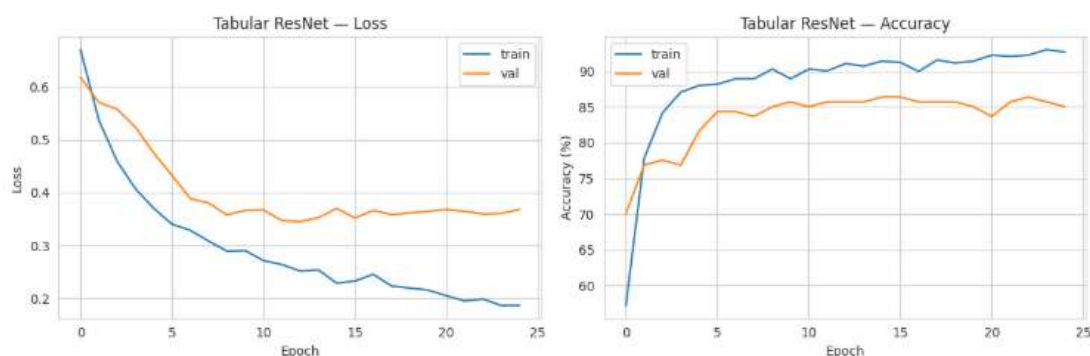


Figure 32 Tabular ResNet Training and Validation Curves with significant feature subset

Figure 33 shows its confusion matrix, which is almost identical to the complete feature version and confirms that feature selection changed the ranking quality of the model more than it changed the raw counts.

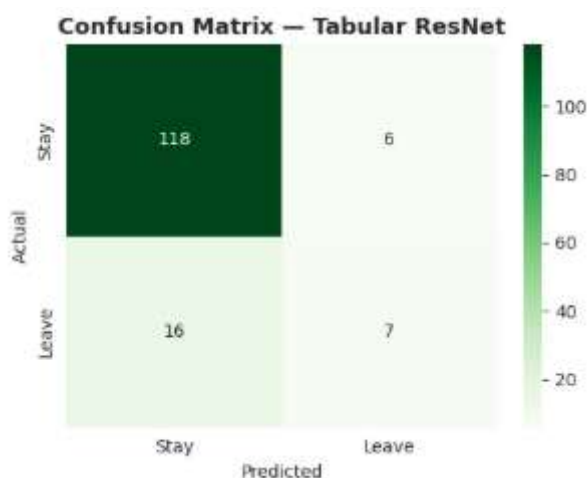


Figure 33 Confusion matrix for the Tabular ResNet with the significant feature subset

As Table 15 reports, the Tabular ResNet retained the same test accuracy of 85.03%, but its ROC-AUC improved from 0.740 to 0.812. Feature selection therefore improved the ability of the model to rank employees by risk even though its accuracy at the default decision threshold was unchanged.

Table 15 Evaluation results of the Tabular ResNet model with the significant feature subset

Class or Metric	Precision or Value	Recall	F1 Score	Support
Test accuracy	85.03%	--	--	--
ROC-AUC	0.812	--	--	--
Stay	0.88	0.95	0.91	124
Leave	0.54	0.30	0.39	23
Macro average	0.71	0.63	0.65	147
Weighted average	0.83	0.85	0.83	147

Figure 34 compares the performance of the FT-Transformer and the Tabular ResNet on the significant feature subset across Accuracy, Precision, Recall, F1 score and ROC-AUC. Read together with Table 14 and Table 15, the chart shows that the FT-Transformer leads on accuracy and on precision for the Leave class, at 0.75 against 0.54, while the Tabular ResNet achieves

higher recall on that class, at 0.30 against 0.26, and a higher ROC-AUC. Their F1 scores on the Leave class are identical at 0.39, so the choice between them depends on whether an organisation would rather flag fewer leavers with greater confidence or catch slightly more of them at the cost of more false alarms.

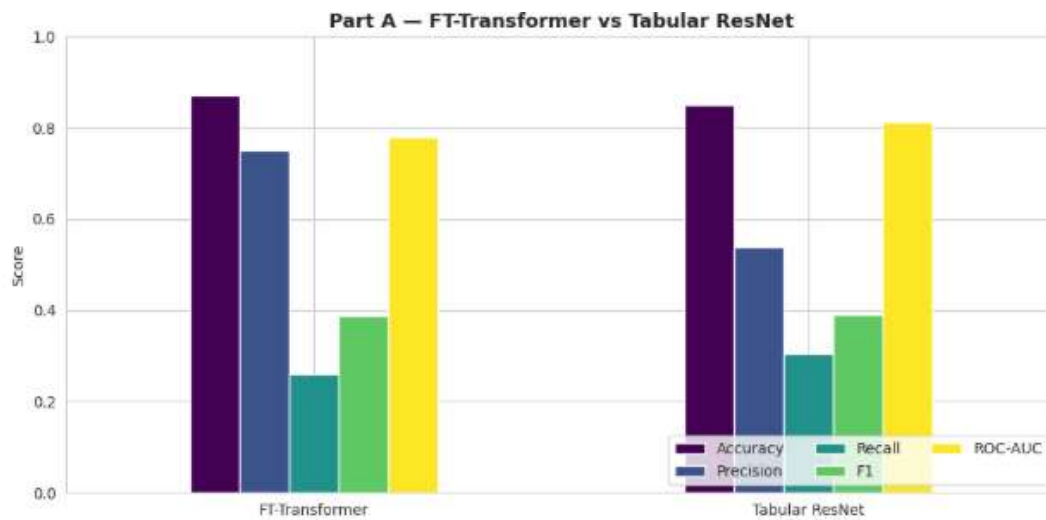


Figure 34 Comparison of the FT-Transformer and the Tabular ResNet on the significant feature subset

4.4 Sentiment Results with the Significant Configuration

For completeness, the sentiment models were evaluated again under the corresponding configuration. Figure 35 shows the BiLSTM training curves, which again rise smoothly before flattening.



Figure 35 BiLSTM training and validation curves for the significant configuration

Figure 36 shows its confusion matrix, which once more splits its few errors fairly evenly between the two classes.

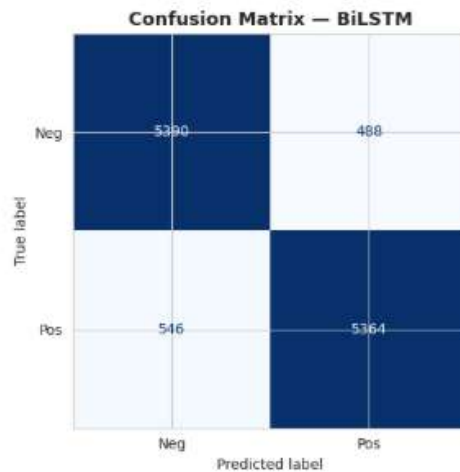


Figure 36 Confusion matrix for the BiLSTM with the significant configuration

As Table 16 reports, the BiLSTM reached a test accuracy of 91.23%, a little below its complete dataset result, with a larger gap between training and test accuracy that points to mild overfitting.

Table 16 Evaluation results of the BiLSTM sentiment model with the significant configuration

Class or Metric	Precision or Value	Recall	F1 Score	Support
Training accuracy	96.33%	--	--	--
Validation accuracy	91.47%	--	--	--
Test accuracy	91.23%	--	--	--
Negative	0.908	0.917	0.912	5,878
Positive	0.917	0.908	0.912	5,910
Macro average	0.912	0.912	0.912	11,788

Figure 37 shows the word-level SHAP explanation for the model. The word “lack” has the strongest influence on the prediction, followed by “stable”, “potentially”, and “successful.” Words such as “people” and “Risk” also contribute to the model's decision, while “rather”, “sucks”, and “risky” have a smaller impact. These results suggest that the model relies primarily on a few key words when making predictions, with the remaining features contributing less overall.

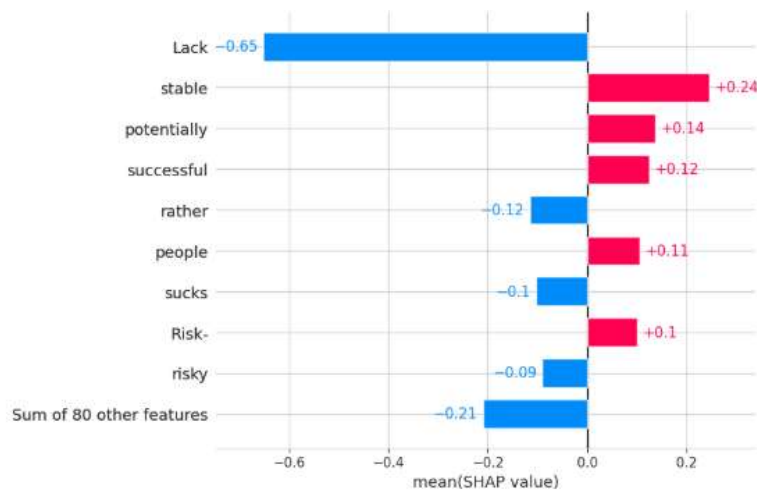


Figure 37 Word level SHAP explanation for the BiLSTM with the significant configuration

Figure 38 shows the CNN-BiGRU training curves for the same configuration.

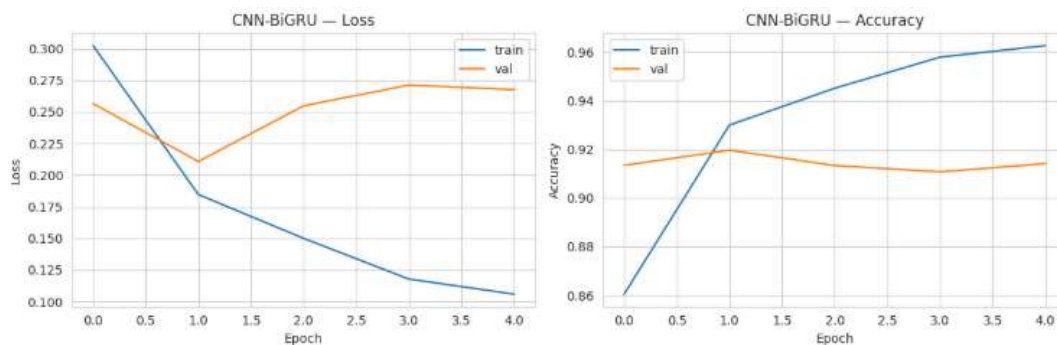


Figure 38 CNN-BiGRU with attention training and validation curves for the significant configuration

Figure 39 shows its confusion matrix, which keeps the strong diagonal seen earlier.

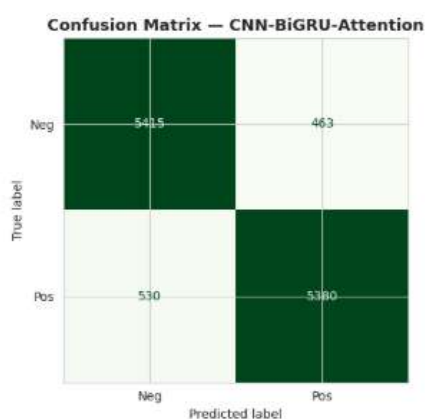


Figure 39 Confusion matrix for the CNN-BiGRU with attention model with the significant configuration

As Table 17 reports, the CNN-BiGRU model reached 91.58% with a ROC-AUC of 0.972, again marginally ahead of the BiLSTM.

Table 17 Evaluation results of the CNN-BiGRU with attention model with the significant configuration

Class or Metric	Precision or Value	Recall	F1 Score	Support
Test accuracy	91.58%	--	--	--
ROC-AUC	0.972	--	--	--
Negative	0.911	0.921	0.916	5,878
Positive	0.921	0.910	0.916	5,910
Macro average	0.916	0.916	0.916	11,788

Figure 40 compares the performance of the BiLSTM and CNN-BiGRU-Attention models using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Both models achieve strong results across all evaluation metrics, with scores above 0.90. The CNN-BiGRU-Attention model shows slightly better performance than the BiLSTM model, particularly in terms of ROC-AUC, indicating a marginally stronger ability to distinguish between classes. Overall, the results suggest that both models are highly effective, although the CNN-BiGRU-Attention model performs slightly better overall.

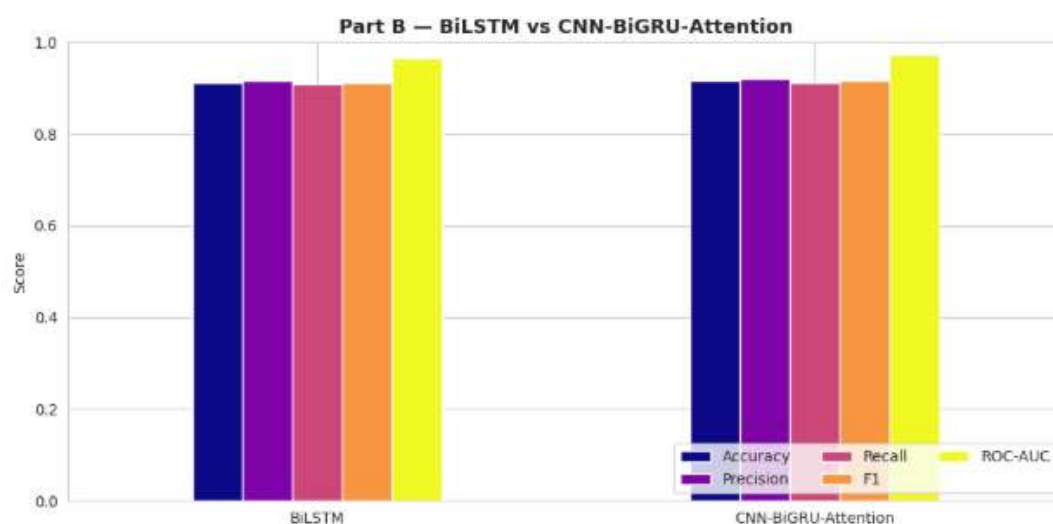


Figure 40 Performance Comparison of BiLSTM and CNN-BiGRU-Attention Models

4.5 Cross Dataset Validation

The models were validated on independent data sets other than the initial original dataset of IBM Employee Attrition to assess generalisability of the proposed framework. The extra evaluation was carried out to determine whether the models were able to continue to be predictive with unseen data, and to make sure results were not affected by overfitting or the specific properties of the data set. The attrition prediction models were evaluated on a Job Change dataset with 19,158 employee records and 12 predictor variables and sentiment analysis models were evaluated on an Employee Reviews dataset containing 67,529 reviews. All four models were evaluated on these sets and the results are summarised in Table 18.

Table 18 Consolidated cross dataset performance of all four models

Model	Dataset and shape	Features	Acc %	Prec %	Rec %	F1 %	ROC-AUC	Params
FT-Transformer	Job Change (19,158 x 14)	12 (all)	79.54	81.00	80.00	80.00	-	260,354
Tabular ResNet	Job Change (19,158 x 14)	12 (all)	79.12	80.00	79.00	80.00	0.812	266,753
BiLSTM	Employee Reviews (67,529 x 19)	Text	91.45	89.25	94.30	91.71	0.971	2,704,450
CNN-BiGRU with attention	Employee Reviews (67,529 x 19)	Text	91.53	89.37	94.31	91.78	0.976	2,723,586

The results demonstrate that both attrition models performed similarly on the Job Change dataset, despite the fact that the employees in this dataset are different than those of IBM and data is distributed differently between these two datasets. This means that the models were able to capture generalizable patterns, not just the relationships present in the training data. In addition, the validation on an independent dataset adds evidence that the models are still solid with variations in workforce compositions and class imbalance.

The sentiment analysis models also performed well with respect to generalisation with an accuracy of above 91% and ROC-AUC values of more than 0.97 for Employee Reviews dataset. These findings validate that the models can retain their high predictive accuracy when the number of data points is increased and the data is obtained from various sources.

For the Job Change dataset and Employee Reviews, variables were available and all of them were found to be significant. Therefore, no additional feature selection was needed and the total features set was used for the training of the models and their evaluation. In general, the cross-dataset results validate the proposed FT-Transformer, Tabular ResNet, BiLSTM and CNN-BiGRU-Attention models as having stable performance across datasets with varying characteristics, sizes and class imbalance across the datasets.

Overall, the cross-dataset validation results validate the effectiveness of the proposed FT-Transformer, Tabular ResNet, BiLSTM, and CNN-BiGRU-Attention models for extending their generalisation capability beyond the IBM and Employee Reviews data set, while keeping performance stable across the different data sets with varying characteristics, size, and class imbalance.

4.6 Deployment

The trained models were placed behind a single web application built with Gradio and hosted on Hugging Face Spaces under the Space name HR-Insights-Attrition-Sentiment, so that the work is not only a set of notebooks but something an HR user can open in a browser and try. The application is organised into two tabs, one for attrition prediction and one for sentiment analysis, and in each tab the user first chooses which of the two trained architectures should make the prediction. Four models are therefore live in total, the FT-Transformer and the Tabular ResNet on the attrition side and the BiLSTM and the CNN-BiGRU-Attention network on the sentiment side. Exposing both architectures rather than only the stronger one was a deliberate choice, since the comparison between them is central to this study and allows the same input to be scored by either model and the difference observed directly. Every prediction is returned with a plainly worded verdict and a feature level or word level explanation, so that the system shows its reasoning rather than only its output.

The attrition tab is served by the models trained on the Job Change dataset, and for the Sentiment tab is served by the Employee dataset Reviews. This dataset was selected for deployment because its fields are the kind of information an employer can realistically collect at the point of assessment, and because the IBM Attrition dataset carried the class imbalance discussed in Section 4.1 and 4.2, which made its accuracy figures a weaker basis for a system intended for live use. The models trained on the IBM Attrition dataset were retained as an experimental comparison and were not deployed. Figure 41 shows the attrition tab in use. The input form is laid out across two columns, with location, background and education on the left

and discipline, experience and company details on the right. Free numeric quantities such as the city development index and training hours use sliders with the valid range shown, while the remaining fields use dropdown menus, which keeps the form quick to complete and prevents invalid entries. In the example, a male graduate with a STEM discipline, ten years of experience and relevant prior experience at a private company of fifty to ninety nine employees is scored by the FT-Transformer.

Real-Time Explainable Deep Learning Platform
Employee Attrition Prediction and Sentiment Analysis.

Attrition Prediction | Sentiment Analysis

Model
☒ FT-Transformer
☐ Tabular ResNet

City: city_103
 City development index: 0.02
 Gender: Male
 Relevant experience: Has relevant experience
 Enrolled university: no_enrollment
 Education level: Graduate

Major discipline: STEM
 Experience (yrs): 10
 Company size: 50-99
 Company type: Pvt Ltd
 Years since last new job: 1
 Training hours: 40

Predict attrition

LIKELY TO STAY WITH THE COMPANY
 The model places this employee below the current decision threshold.
94.1% CONFIDENCE
5.9% PROBABILITY OF LEAVING
-3.5 pts
LOW RISK RISK RATING
Decision threshold: 0.075. Average employee in the training data scores 0.45. Model: FT-Transformer

Prediction probabilities

PUSHING TOWARDS LEAVING			PUSHING TOWARDS STAYING		
FEATURE	VALUE	INFLUENCE	FEATURE	VALUE	INFLUENCE
Education Level	Graduate	0.45	Years Since Last Job Change	1	2.98
Training Hours	40	0.30	Years of Experience	10	0.46
Gender	Male	0.07	City Development Index	0.02	0.14
Company Size	50-99	0.05	University Enrollment	no_enrollment	0.01
City	city_103	0.07	Relevant Experience	Has relevant experience	0.12
Major Discipline	STEM	0.03	Company Type	Pvt Ltd	0.03

Influences measured by modifying one feature to a low (leaving) or to the opposite (staying) value and recording how the model's output changes. 1 point below zero the feature moved the decision further towards leaving relative to the employee's fit.

Figure 41 Deployed attrition prediction tab on Hugging Face Spaces

The result appears directly beneath the form. A headline verdict states that the employee is likely to stay with the company, supported by four figures, a confidence of 94.1%, a probability of leaving of 5.9%, a comparison of minus 3.5 points against the average employee in the training data, and a summary risk rating of low. The decision threshold and the training data average of 9.4% are printed alongside so that the reader can see what the verdict is being measured against, together with the name of the model that produced it. Below this, the explanation is split into two tables. One lists the features pushing the prediction towards leaving and the other lists the features pushing it towards staying, each with the value the user entered and a bar showing the size of that feature's effect. These influences are calculated by masking each of the features and setting it back to the training average and seeing how far the output of the model shifts. Bigger bars are features that moved the decision further. These values are relative to that employee only. For the example shown, years since last job change and years of experience were the most influential factors keeping the prediction towards staying, while education level and training hours were the most influential factors pushing it towards leaving.

The second tab is employee sentiment and follows the same pattern. The user can select the BiLSTM or the CNN-BiGRU-Attention model and paste a review or a piece of free text feedback in the input box and submit it. The BiLSTM classifies a short review as shown in Figure 42. The verdict panel reports a positive review with a confidence of 99.2%, a positive probability of 99.2%, a gain of 64.1 points relative to an empty review and a strength rating of strong, with the threshold, the number of words analysed against the cap of eighty and the model's name printed underneath.

The full probability split across the two classes is available in a collapsible panel. The explanation is then given at word level. The review is redisplayed with each word shaded according to whether it pushes the sentiment positive or negative, and the two tables beneath list the contributing words with the size of each effect. In this example the word great carried almost all of the positive reading, with pay contributing a smaller amount and no words pushing the review negative. As on the attrition tab, these effects are found by masking each word in turn and observing how the output moves, and stopwords are removed and the remaining words lemmatised before the model sees them.

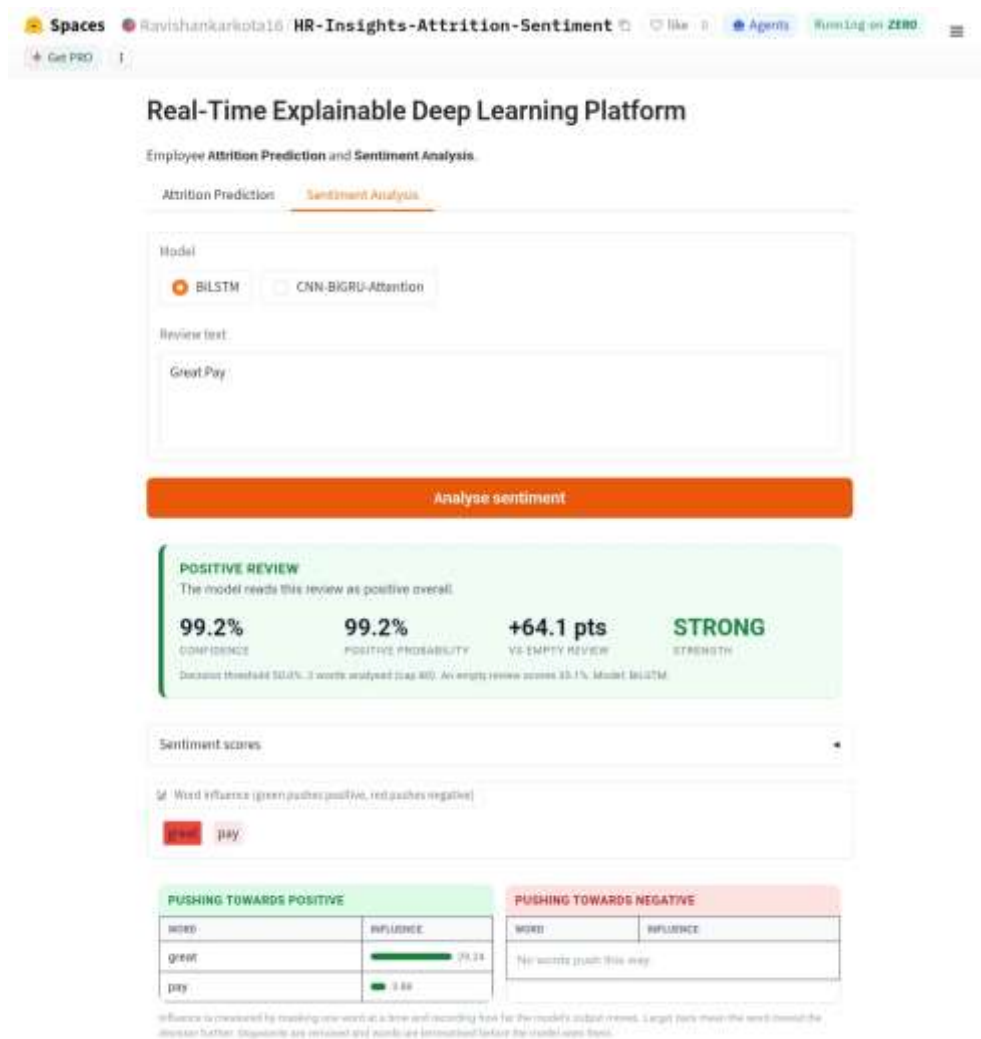


Figure 42 Deployed sentiment analysis tab on Hugging Face Spaces

Taken together, the two tabs meet the deployment aim set out at the start of the project. Both prediction tasks are reachable from an ordinary web page, both architectures for each task are live rather than only the best scoring one, and each result is returned with a plainly worded verdict and an explanation of what drove it, which is what makes the system usable by a reader without a machine learning background.

4.7 Related Works

Table 19 sets the results of this project against the studies reviewed in Chapter 2. On the attrition task the FT-Transformer reached 87.76% on the test set, which is slightly ahead of the strongest traditional results in the literature, the 87.46% Random Forest of Rohit et al. and the 87.41%

Random Forest of Inggrianti and Mulyana, and well ahead of the 83% CNN-LSTM of Ramya and Sanjay. The gap in accuracy is small and would not on its own justify the extra cost of a transformer. What the present work adds is the explanation layer. None of the studies above went beyond built-in feature importance, whereas this project pairs the model with SHAP, attention weights and permutation importance, so a prediction can be traced back to the features that produced it rather than only reported.

On the sentiment task the CNN-BiGRU model with attention reached 91.79% on real employee reviews. That sits close to the 92.36% Rahman et al. reported on IMDb and above the 90% Reddy et al. reported, but both of those results come from film reviews and social media posts. Workplace feedback has had far less attention, and the figures here show that architectures developed on general review text carry over to it without loss.

Table 19 Comparison of the project results with related work

Study	Approach	Dataset	Best accuracy	Explainability
Krishna et al. [1]	Random Forest, KNN, Naive Bayes, ensemble	IBM HR	84%	Random Forest importance
Rohit et al. [2]	Random Forest	IBM HR	87.46%	Random Forest importance
Inggrianti and Mulyana [3]	Random Forest with web app	IBM HR	87.41%	Random Forest importance
Ramya and Sanjay [4]	CNN-LSTM hybrid	IBM HR	83%	None
Pathak et al. [5]	Logic Tensor Network and XGBoost	IBM HR	86.2%	Built in rules
This project (attrition)	FT-Transformer	IBM HR/ Job Change	87.76% and 79.54%	SHAP, attention, permutation
Rahman et al. [7]	RoBERTa-BiLSTM	IMDb and Twitter	92.36% and 80.74%	Not reported
Reddy et al. [6]	BiLSTM and DistilBERT-GRU	Movie and social media	90%	LIME
This project (sentiment)	CNN-BiGRU with attention	Glassdoor/ Employee dataset Reviews	91.79% and 91.53	LIME, SHAP

Two limitations sit against these numbers. Minority class recall on the IBM dataset stayed low for both attrition models, so the system is better at confirming who will stay than at catching everyone who will leave. With 1,470 records and a leaver rate near 16%, that follows from the data rather than the architecture, and no change of model was going to fix it. This is why the IBM models were kept as an experimental comparison and were not put into the deployed application. A model that misses leavers is the wrong thing to hand to an HR user, since the leavers are the cases the tool exists to find, and a headline accuracy of 87.76% would give a false sense of how well it was doing. The attrition tab therefore runs on the Job Change models, which were trained and validated on a larger and better balanced dataset, so the predictions returned to the user are not skewed towards the majority class. The sentiment models were also trained on positive and negative reviews only, with the neutral class dropped, and a three class version would be more useful in practice since a good deal of real feedback is neither. Both points feed into the future work set out in the next chapter.

Chapter 5 Conclusion

This project set out to build a real time explainable deep learning platform that predicts employee attrition, reads the sentiment of employee reviews, and explains both inside one deployable system. The objectives set out in Chapter 1 were met. An FT-Transformer and a Tabular ResNet were developed for attrition prediction, a BiLSTM and a CNN-BiGRU model with attention were developed for sentiment analysis, all four were compared under identical training conditions on Accuracy, Precision, Recall, F1 score and ROC-AUC, and all four were explained and then deployed.

The first finding is that statistical feature selection changed very little. Test accuracy for the FT-Transformer moved from 87.76% to 87.07% and the Tabular ResNet held at 85.03% in both settings, which says the models are robust to the feature configuration they are given. What feature selection did change was the quality of the ranking rather than the count of correct answers, the ROC-AUC of the Tabular ResNet rose from 0.740 to 0.812, and the precision of the FT-Transformer on the Leave class rose from 0.69 to 0.75, though its recall on that class fell from 0.39 to 0.26. The reduced models are cheaper to run and easier to explain, and they lose nothing in accuracy, but the 21 significant features were not carrying information the 30 features did not already hold.

The second finding is the one that shaped the rest of the project. Overall accuracy near 88% looks strong until the confusion matrix is read, at which point it is clear that both attrition models identify stayers almost perfectly, at a recall of 0.97 for the FT-Transformer, and miss most leavers. With 1,470 records and a leaver rate near 16%, this follows from the dataset rather than the architecture, and neither class weighting nor threshold adjustment removed it. This is why the IBM models were kept as an experimental comparison and were not put into the live application. A tool that misses the leavers is the wrong thing to give an HR user, since the leavers are the reason the tool exists. The attrition tab therefore runs on the Job Change models, which were trained on a larger and better balanced dataset in which all 12 predictors tested as significant, and which reached accuracies close to 80% on data the models had never seen.

The third finding comes from the explainability work. OverTime was the only feature to rank highly under all three methods, SHAP, FT-Transformer attention and Tabular ResNet permutation importance, which makes it the most reliable single signal in the data, employees working overtime carried a clearly higher predicted probability of leaving. Beneath it the methods diverged, with attention favouring Department, MonthlyIncome and

EnvironmentSatisfaction while permutation importance favoured YearsSinceLastPromotion and Age, and the attention weights were spread narrowly between roughly 0.02 and 0.06, which shows the FT-Transformer draws on many features rather than leaning on a few. When taken together, these picture cues point to workload, pay, working environment and lack of progress as the levers that impact retention. The sentiment models agreed with each other, but from a different perspective, words like lack, stressful and bureaucratic had the strongest negative sentiment, and stable was the strongest positive indicator and indeed the models picked up on the vocabulary of the workplace, not the generic sentiment.

The final product is the deployed system. Four models are deployed in a single gradio app on huggingface spaces, each for a different task, with the architecture choice being shown, so that the user could repeat the experiment presented here, and not just read about it. The attrition tab gives a verdict, a confidence, a probability of leaving, and a comparison to the training average with two tables showing the features that are pushing the result either way. The sentiment tab displays the class, the confidence, and the review word by word for which the class was moved by. The above explanations are generated by masking one feature or word at a time and observing the shift in output that is, the explanation presented is the model's own and is not a summary of it. Overall, the project provides an integrated and explainable platform for HR decision support applications that integrates the prediction of attrition with sentiment analysis into a single deployable system, along with a transparent indication of where the predictions can be trusted and where they cannot.

5.1 Future Work

The most immediate need is the imbalanced class which restricted the prediction of attrition throughout this project. Both models had performed well at identifying stayers but failed to capture most leavers and improving their ability to do this will need more comprehensive HR data and a more cost sensitive learning and threshold optimisation process, to ensure the models focus on the most important leavers rather than on accuracy. On the sentiment side, there are ways to expand the models from binary to three class classification simply by adding back the neutral reviews that were not here, as a significant portion of workplace feedback is in the middle of the spectrum. It is also possible to consider stronger transformer based language models, like RoBERTa, DeBERTa or a variant of BERT specific to the application domain, but increasing the accuracy of such models would come with the cost of slower response in an interactive environment.

The next step is to shift the platform from static to real-time HR data, enabling predictions of attrition and sentiment to evolve over time as new employee data and reviews enter the system, instead of having to be retrained. The use of open public datasets provides a static workforce that would be vastly improved with either licensed or consented workforce data, managed according to GDPR and under data sharing agreements in a real organisation. The two pipelines could then be concatenated instead of operating side by side, and the sentiment score of the comment made by the employee could be fed into the attrition model as an additional feature, for opinion and structured record to feed into a single prediction. Predictions and explanations would then be automatically generated as employee information changed, and would be integrated into an existing Human Resource Management or Employee Information System and supported with incremental and online learning.

Finally, the platform could be developed from a tool that predicts into one that also advises. The explanation outputs already identify the factors behind each decision, so these could be turned into personalised retention recommendations such as workload adjustments, development opportunities or compensation reviews. Evaluating the system across organisations in different industries and countries would show whether the drivers found here, particularly the dominance of OverTime, generalise beyond this dataset or are specific to it. The most important test is human usability study with HR managers would establish whether the explanations produced by SHAP, LIME and the attention mechanisms are genuinely understandable and trusted by the people meant to act on them, which has been assumed throughout this project but never directly tested.

5.2 Reflection

Looking back, this project taught me more about judgement than about coding. I went in thinking the models would be the hard part, but they weren't. The real work was figuring out what the results actually meant, and being honest about where the system could be trusted and where it couldn't.

The turning point was the confusion matrix. Nearly 88% accuracy looked great, and I could easily have stopped there and been happy with it. But when I looked properly, the models were spotting the people who stayed and missing most of the ones who left, and the leavers are the whole reason the tool exists. So that nice looking number was actually hiding a failure. That taught me a single metric can lie to you, and that class imbalance isn't a small detail. More than

anything it forced me to think critically, to stop trusting a result just because it looked good and start asking what was really sitting behind the number.

I tried class weighting and threshold tuning to fix it, and honestly I was frustrated when neither really worked. The problem was the data, not the model, since only about 16% of the IBM records were leavers. Accepting that was hard, but it led to my biggest decision, which was keeping the IBM models as a comparison only and running the live tool on the better balanced Job Change data instead. I learned it is better to follow the evidence than to defend a choice you have already made.

Feature selection caught me off guard too. I expected it to clearly help or hurt accuracy, but it barely moved it, while still making the models cheaper to run and easier to explain. So I learned that nothing changing can be a real result in itself.

The explainability side was my favourite part. Watching OverTime come out on top across every method told me the models were picking up real patterns, not noise. It also brought home why explanation matters when the prediction is about an actual person, and getting the system to explain each result felt like the most meaningful thing I built.

Deployment was harder than I expected. Getting four models to run together in one app on Hugging Face Spaces meant grinding through a lot of configuration and version issues. It was frustrating at the time, but it showed me the real distance between a model that works in a notebook and one that works as a product.

If I did it again, I would choose a better balanced dataset from the start, plan the explainability earlier instead of bolting it on, and spend more time on the data upfront, because nearly every problem I hit later traced back to something in it.

Taken as a whole, this project changed how I work as a data scientist. It taught me to think critically about every result, to check my findings instead of taking them at face value, and to base decisions on evidence rather than assumptions. It sharpened my skills in deep learning, NLP and deployment, but the bigger lesson was how a data analyst should actually think, looking past the surface, questioning the data, and staying honest about what the numbers can and cannot say. I am finishing more confident in what I can build, and a lot more aware of the responsibility that comes with making predictions about people.

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Appendix A: Project Management

Date	Activity
15 May 2026	Workshop 1: What is Research, Research Questions, Problem Statement, Types of Research
22 May 2026	Workshop 2: Literature Review
29 May 2026	Workshop 3: Methodology, Results and Analysis, and Conclusion
05 Jun 2026	Workshop 4: Responsible Use of AI in the MSc Dissertation Project
08 Jun 2026	Scoping and Literature Review, finalise research problem, objectives, and project plan
15 Jun 2026	Data Acquisition and Exploratory Data Analysis (EDA) on IBM HR Analytics and Glassdoor Review datasets
22 Jun 2026	Data Preprocessing and Feature Engineering with encoding, scaling, tokenisation, and class balancing
29 Jun 2026	Attrition Modelling: FT-Transformer implementation and evaluation
06 Jul 2026	Attrition Modelling: Tabular ResNet implementation, hyperparameter tuning, and comparison
13 Jul 2026	Sentiment Modelling: BiLSTM development and evaluation
20 Jul 2026	Sentiment Modelling: CNN-BiGRU-Attention development and comparative analysis
27 Jul 2026	Cross-Dataset Validation, Explainability Analysis, Deployment Preparation, and Results Consolidation

Appendix B: Artefact/Dataset

Dataset/ Artefact:

1. IBM HR Analytics Employee Attrition & Performance Dataset Link:
<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>
2. Glassdoor Employee Reviews Dataset Link:
<https://www.kaggle.com/datasets/davidgauthier/glassdoor-job-reviews>
3. HR Analytics: Job Change of Data Scientists Dataset Link:
<https://www.kaggle.com/datasets/arashnic/hr-analytics-job-change-of-data-scientists>
4. Employee dataset Reviews Link:
<https://www.kaggle.com/datasets/kunalpatil2181/employee-reviews>
5. HR Analytics: Job Change of Data Scientists and Employee dataset deployed model Link: <https://huggingface.co/spaces/Ravishankarkota16/HR-Insights-Attrition-Sentiment>
6. IBM HR Analytics Employee Attrition & Performance and Glassdoor Employee dataset deployed Model Link:
https://huggingface.co/spaces/Ravishankarkota16/Employee_Attrition_and_Sentiment_Analysis

Github repository link: <https://github.com/kotaravishankar/Real-Time-Explainable-AI-Platform-for-Attrition-Prediction-and-Employee-Sentiment-Analysis>

Appendix C: Use of Artificial Intelligence

AI tools, including Grammarly, were used during the preparation of this report to reduce spelling and grammar errors. ChatGPT was also used to help identify and understand programming errors encountered during code development and to suggest possible debugging approaches. All code, design, implementation, experimentation, analysis, and final decisions described in this report are my own work.